

Bioquímica Estrutural:

Estudo da estrutura das moléculas biológicas e dos princípios que a regem.

(~ **Biologia Estrutural**)

Macromoléculas biológicas

- **DNA:** repositório da informação genética na maioria dos organismos vivos
- **RNA:** transferência (e repositório) de informação genética, matriz para a síntese proteica, funções estruturais, etc...
- **Proteínas:** componentes estruturais (pele, ossos, músculo, cabelo, etc...), catálise de reacções bioquímicas (enzimas), transmissão de sinais, regulação, transdução de energia, etc.,etc., etc.!..
- **Lípidos:** componentes essenciais das membranas biológicas, sinalização
- **Polissacáridos:** armazenamento de energia, função estrutural

Sequência



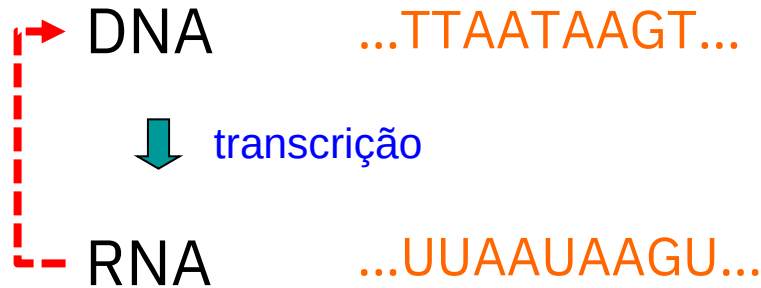
Estrutura



Função

Fluxo de informação biológica

vírus de RNA



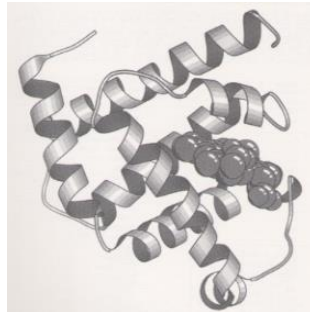
↓ splicing, tradução

cadeia
polipeptídica ...LISVHDN...

↓ modificações pós-translacionais

proteína

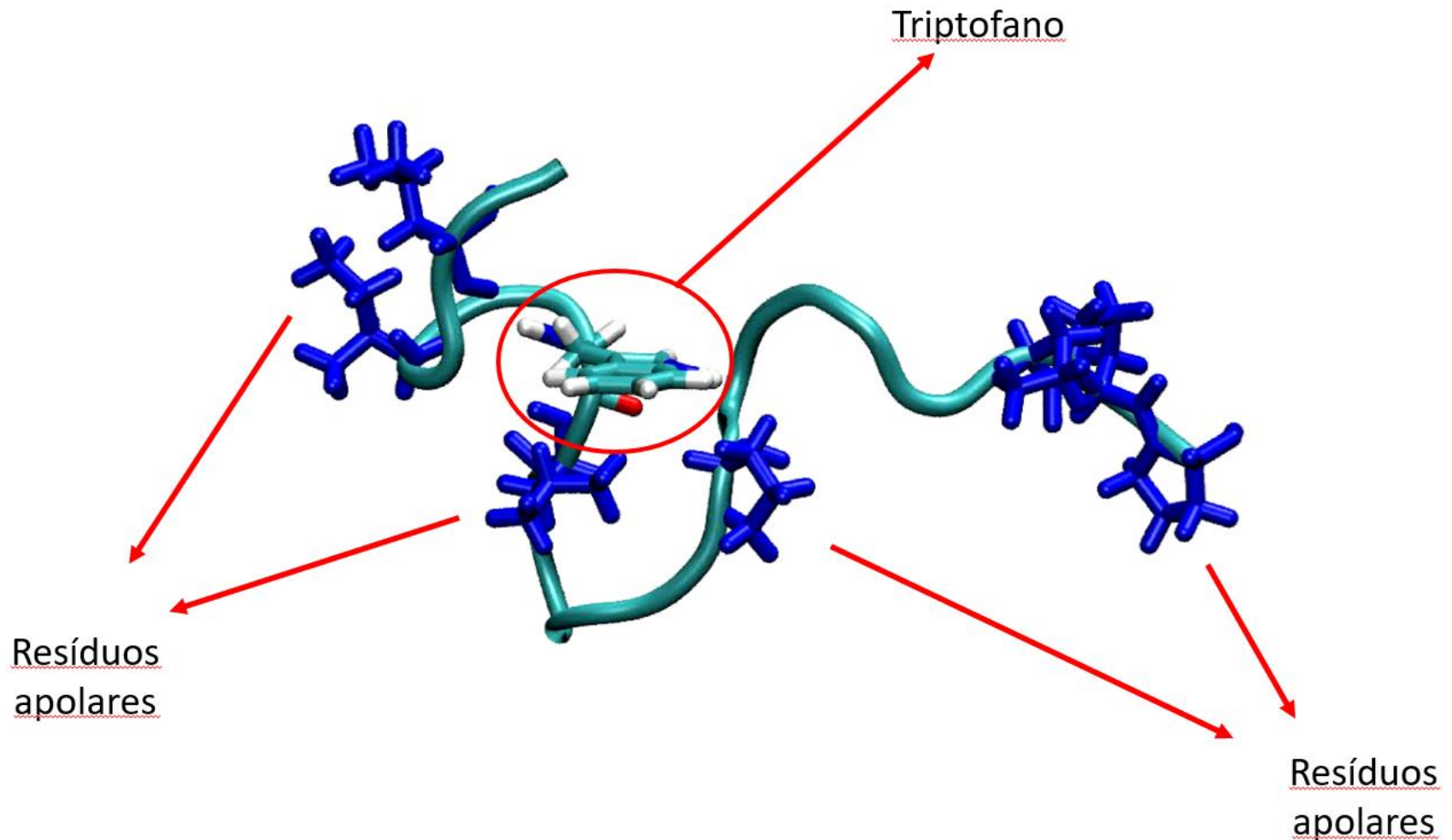
príões



Dogma central da
biologia molecular

Excepções: vírus de RNA,
príões, ribozimas (?)

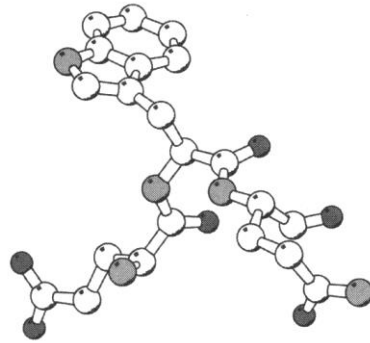
Aquisição espontânea da estrutura tridimensional (Sequência → Estrutura)



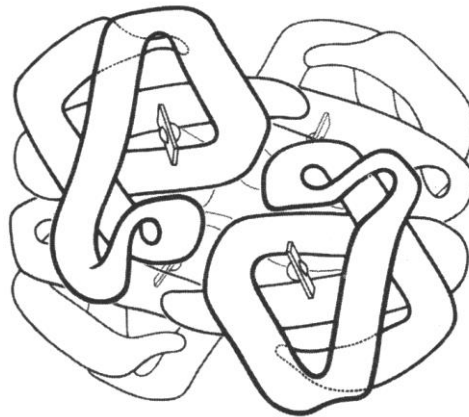
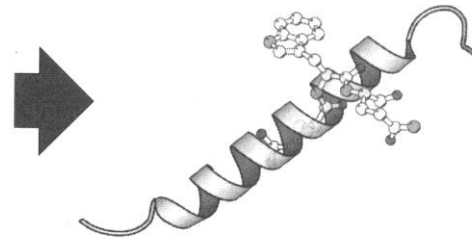
Folding da proteína **Trp-cage** (simulação computacional)

Níveis de organização da estrutura das proteínas

Estrutura primária



Estrutura secundária

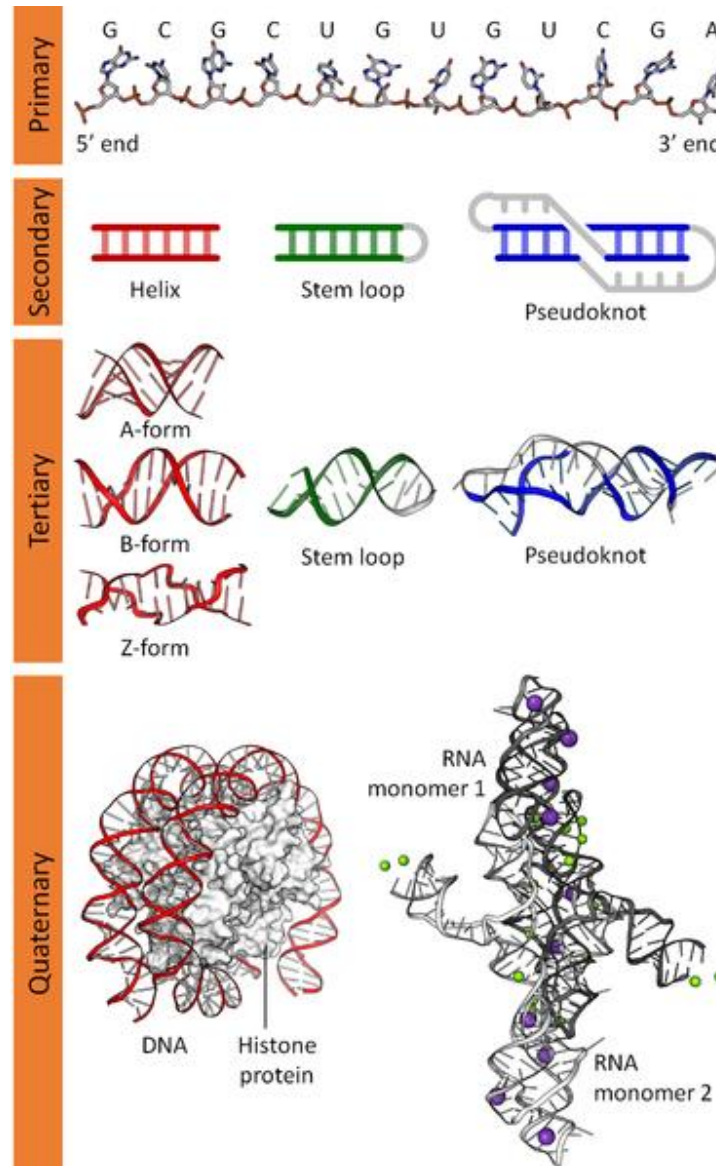


Estrutura quaternária



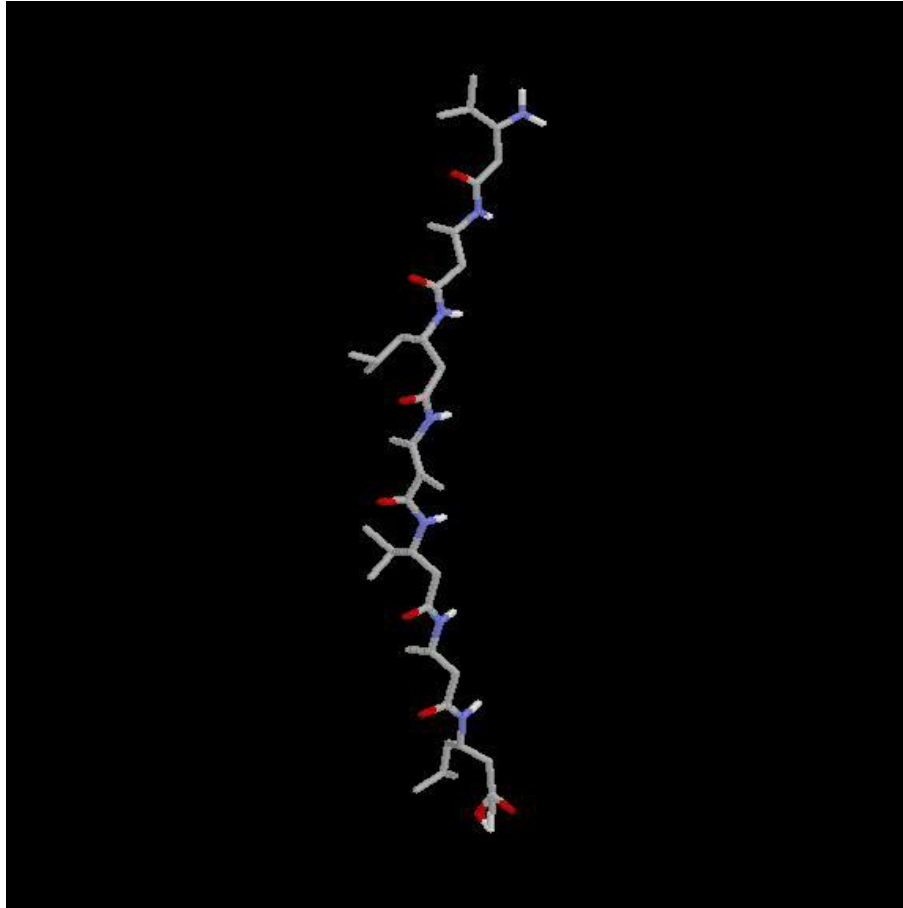
Estrutura terciária

Níveis de organização da estrutura de ácidos nucleicos



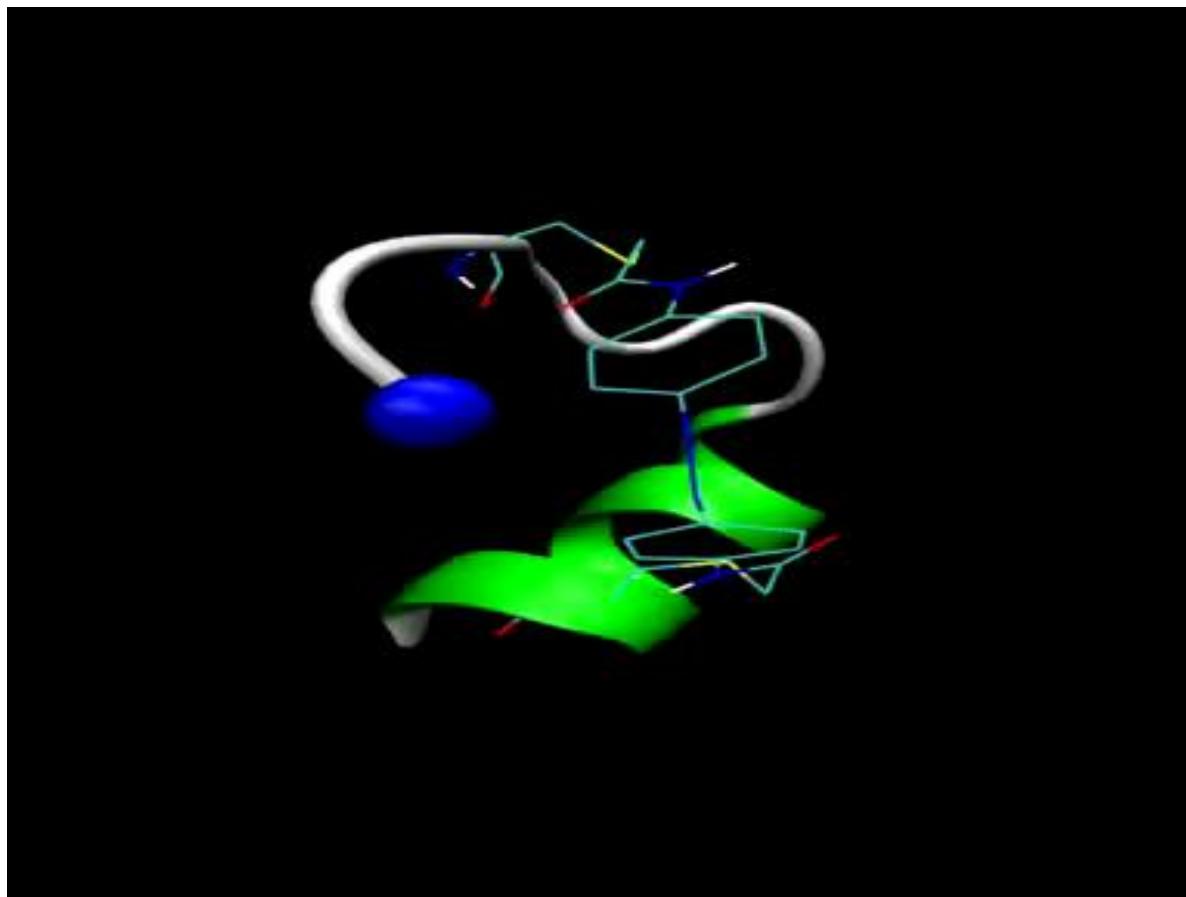
Estrutura quaternária
pode envolver moléculas
de proteína

Formação hierárquica da estrutura



Formação espontânea de uma hélice α
(simulação)

Formação hierárquica da estrutura



Formação espontânea de uma hélice α
(simulação)

De onde provêm a informação estrutural ?

Combinação de vários tipos de conhecimento:

- Teoria da ligação química
- Geometria de moléculas pequenas
- Métodos experimentais para a determinação da estrutura:
 - ❖ Cristalografia de raios X
 - ❖ Ressonância Magnética Nuclear (NMR)
 - ❖ Microscopia Electrónica
 - ❖ Outros métodos (difração de neutrões, SAXS, etc)

Que informação temos disponível ?

(dados do Protein Data Bank em 5/02/2024)

Número de estruturas tridimensionais (coordenadas atómicas):

- Proteínas: 186680
- Ácidos Nucleicos: 4373
- Complexos ácido nucleico-proteína: 12606
- Proteína/Oligossacárido: 11514
- Outros: 228

Total: 215401 estruturas

Métodos experimentais de determinação da estrutura:

- Cristalografia de raios X: 181272 (**84.6%**)
- Microscopia electrónica: 18720 (8.69%)
- NMR em solução aquosa: 14164 (6.58%)
- Métodos híbridos: 229
- Outros: 116

Que informação temos disponível ?

(dados do Protein Data Bank em 3/02/2025)

Número de estruturas tridimensionais (coordenadas atómicas):

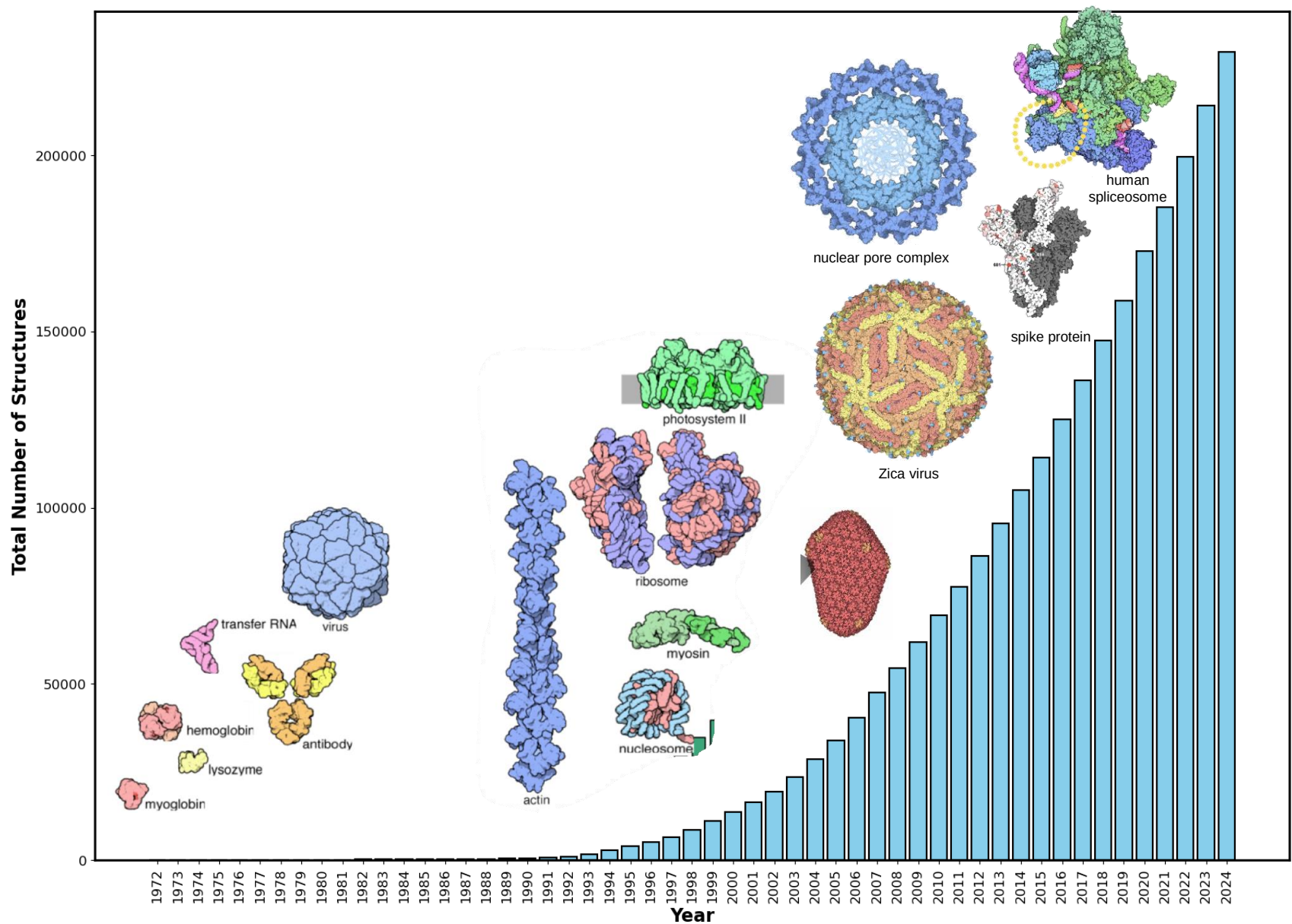
- Proteínas: 199043
- DNA: 2526
- RNA: 1918
- Complexos ácido nucleico-proteína: 14102

Total: 230744 estruturas

Métodos experimentais de determinação da estrutura:

- Cristalografia de raios X: 191248
- Microscopia eletrónica: 24685
- NMR em solução aquosa: 14450
- Métodos híbridos: 238
- Outros: 123

Progresso na determinação das estruturas



Access Computed Structure Models (CSMs) of all available model organisms. [Learn more](#)

Welcome

Deposit

Search

Visualize

Analyze

Download

Learn

RCSB Protein Data Bank (RCSB PDB) enables breakthroughs in science and education by providing access and tools for exploration, visualization, and analysis of:

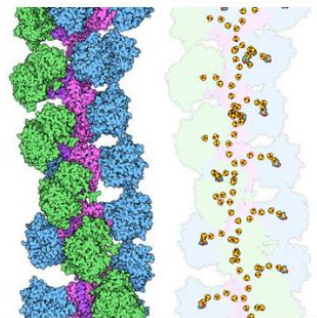
- Experimentally-determined 3D structures from the **Protein Data Bank (PDB)** archive
- Computed Structure Models (CSM)** from AlphaFold DB and ModelArchive

These data can be explored in context of external annotations providing a structural view of biology.

Explore NEW Features

PDB-101 Training Resources

February Molecule of the Month



Nanowires

Latest Entries *As of Tue Jan 30 2024*



8GLK

The Type 9 Secretion System dGldL peak II, NucA substrate bound complex

Features & Highlights

- Register for the February 5 RCSB.org Office Hour**
Have questions about how to use RCSB.org? Join us for a virtual Office Hour.
- Notice: NGL Viewer Deprecation**
As of June 28, 2024, the NGL molecular viewer will be removed from RCSB.org. Users are encouraged to use Mol* for 3D visualization.
- Take the CSM User Survey and Win**
Please take this brief survey about Computed Structure Models at RCSB.org to be entered into a drawing for a set of Bound Playing Cards.

News **Publications**

- February 4 is World Cancer Day**
PDB structures reveal how cell growth is normally controlled, and how cancer cells circumvent these essential controls
» 02/01/2024
- Register for VIZBi 2024 (March 13–15)**
Learn the latest data visualization methods and techniques
» 01/30/2024
- Register for the February 13 Mol* Webinar**
Join RCSB PDB to learn how to Visualize Biomolecular structures with Mol*: From Atoms to Movies
» 01/19/2024

PDB at a Glance 66,758 Structures of Human Sequences 17,179 Nucleic Acid Containing Structures
CSM at a Glance 999,251 AlphaFoldDB 69,326 ModelArchive [More Statistics](#)

Princípios que regem a estrutura das biomoléculas

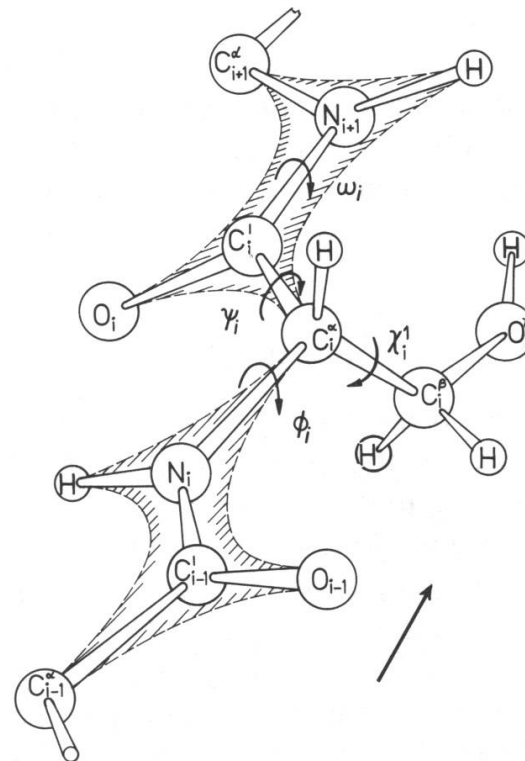
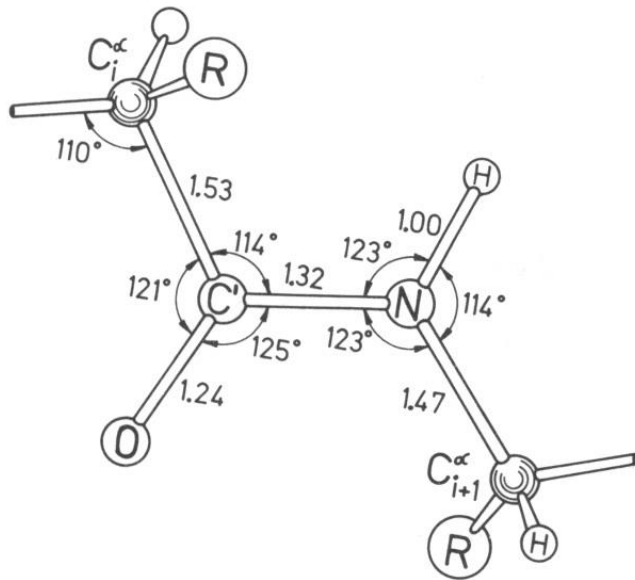
“Perhaps the most remarkable features of the molecule are its *complexity* and *lack of symmetry*. The arrangement seems to be almost totally lacking the kind of regularities which one instinctively anticipates and it is more complicated than has been predicted by any theory of protein structure”

J.C. Kendrew *et al.*, 1958

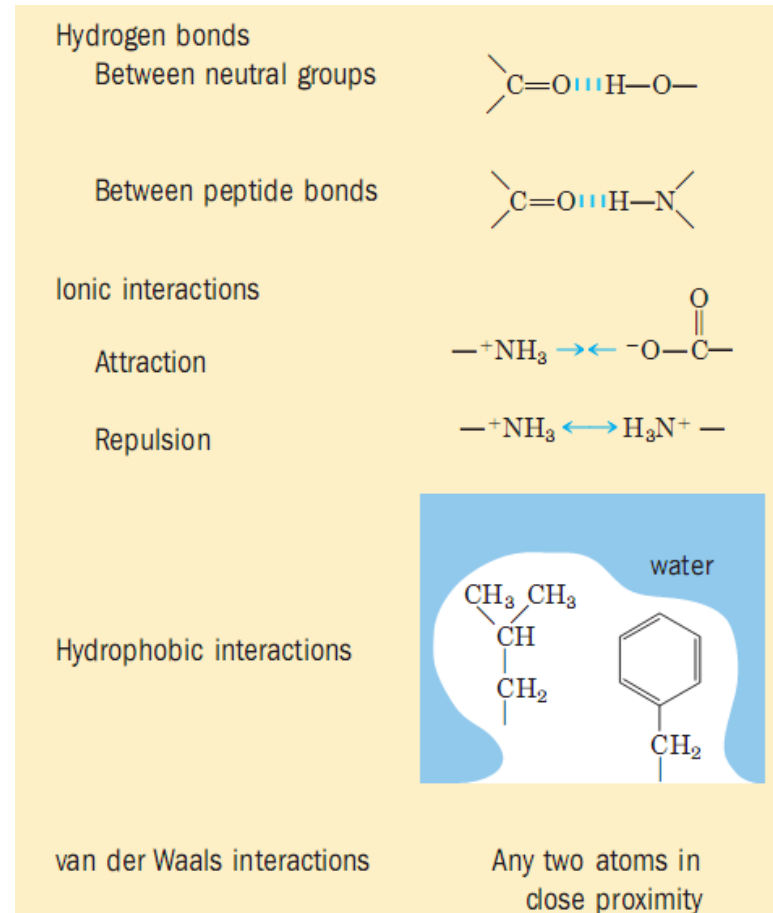


- As macromoléculas biológicas parecem, numa primeira análise, distanciar-se dos princípios simples de geometria e simetria que sabemos reger a estrutura das moléculas pequenas.

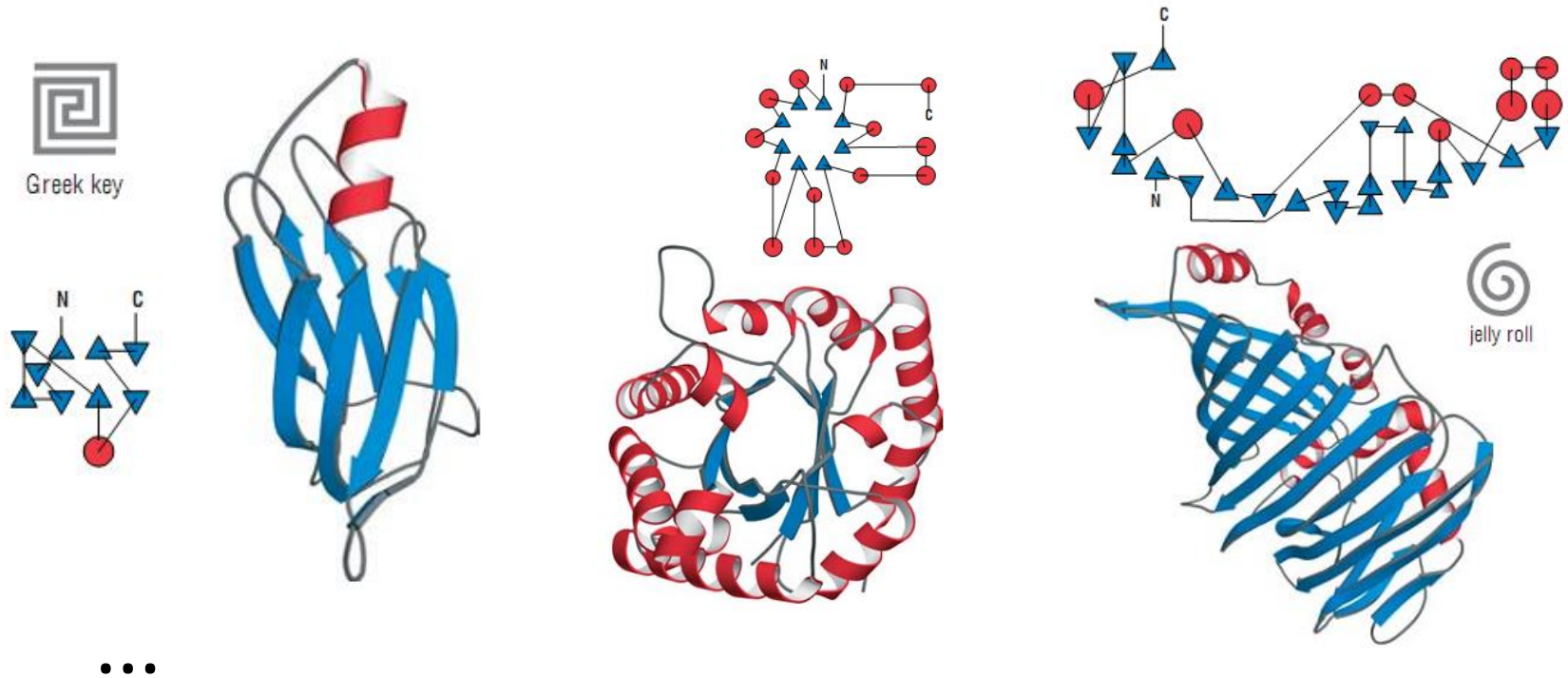
- Ligações covalentes, geometria molecular:



•Interações não-covalentes:

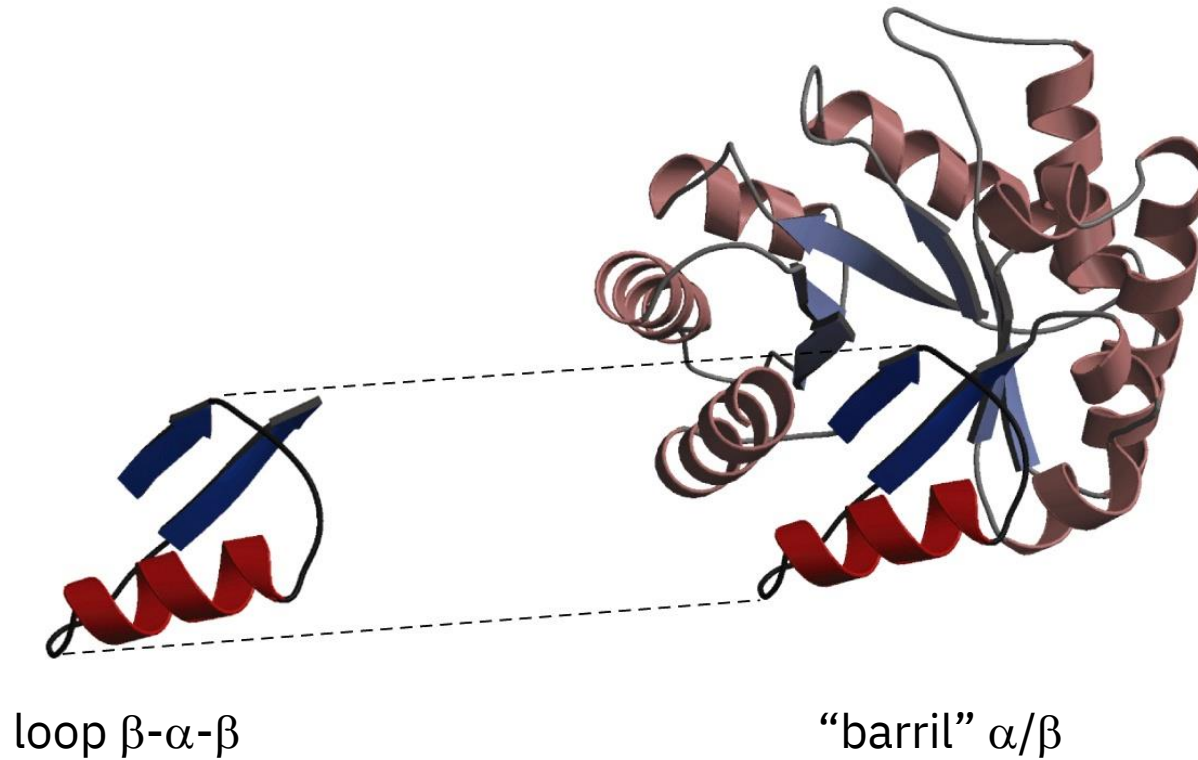


- Princípios arquitetônicos



Recorrência de padrões estruturais na arquitetura das biomoléculas.

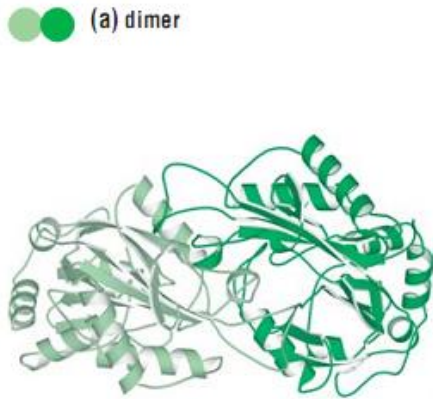
- Princípios arquitectónicos (cont.):



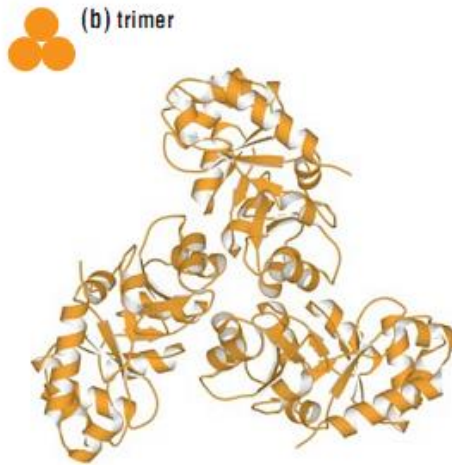
Formação de estruturas a partir da associação de unidades estruturais

- Oligomerização:

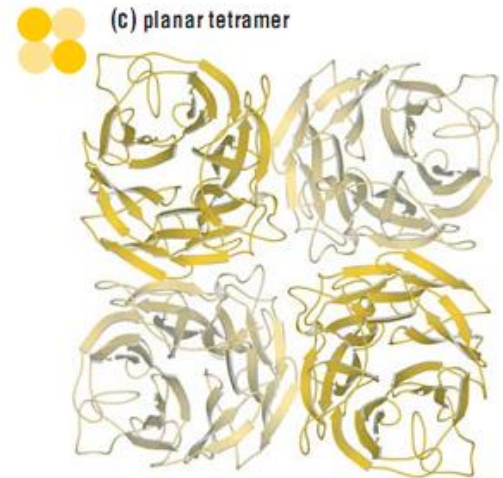
(a) dimer



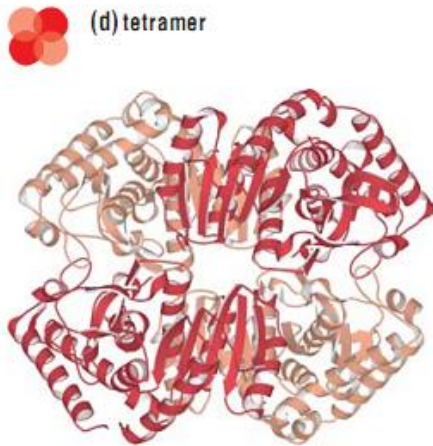
(b) trimer



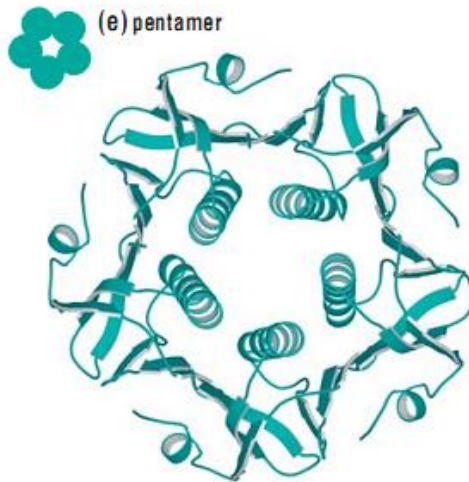
(c) planar tetramer



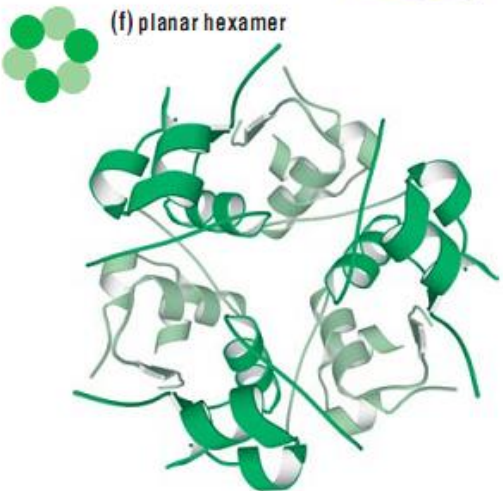
(d) tetramer



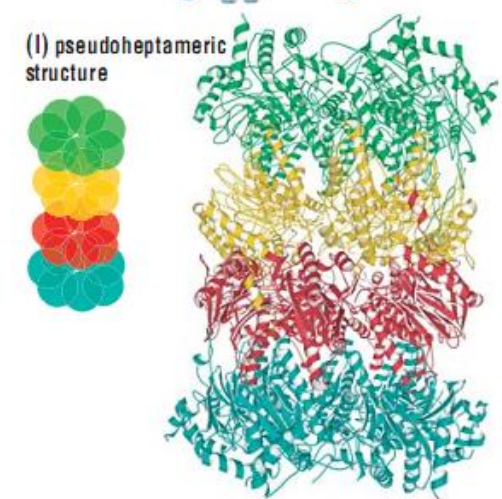
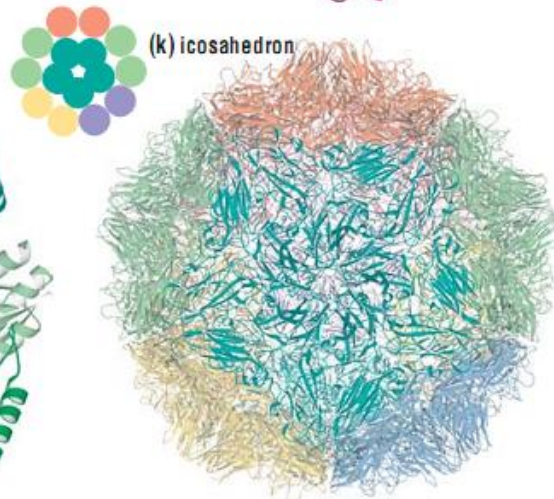
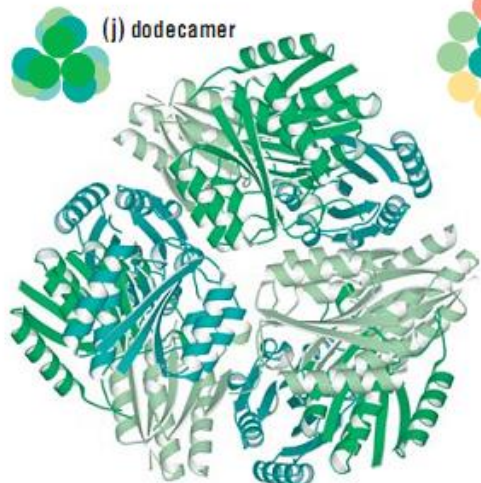
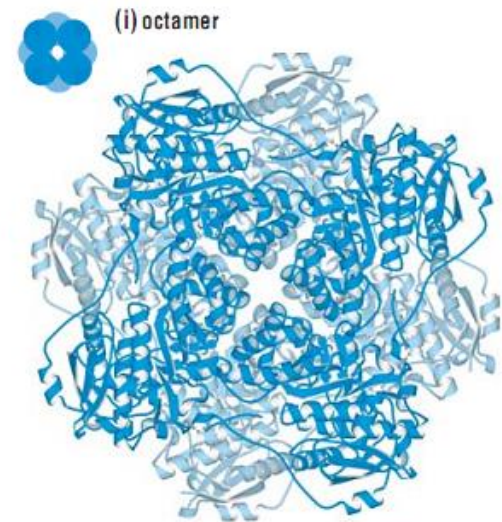
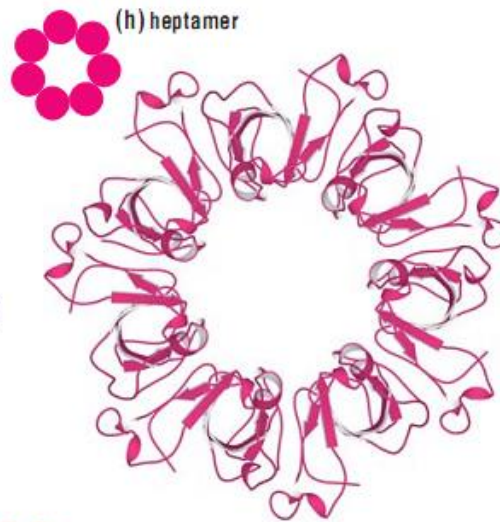
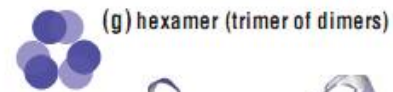
(e) pentamer



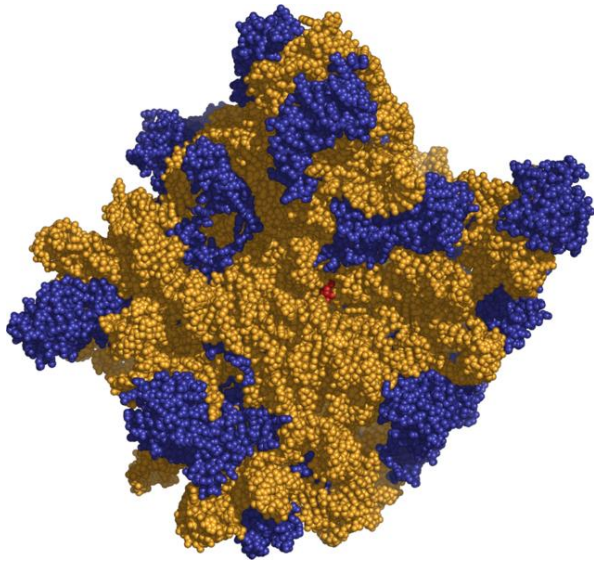
(f) planar hexamer



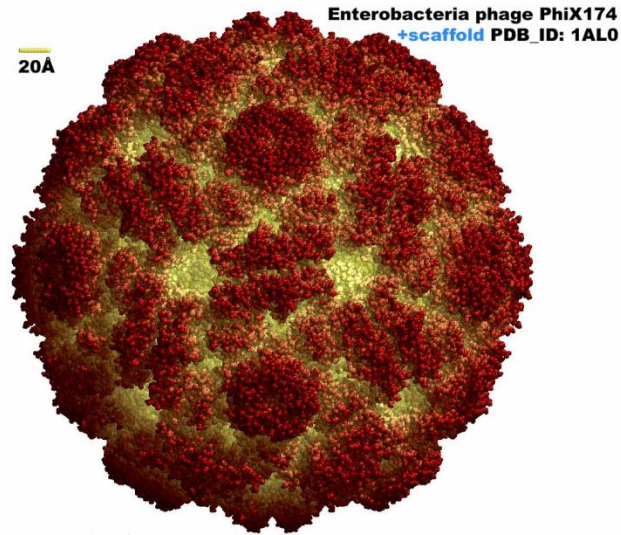
- Oligomerização(cont.):



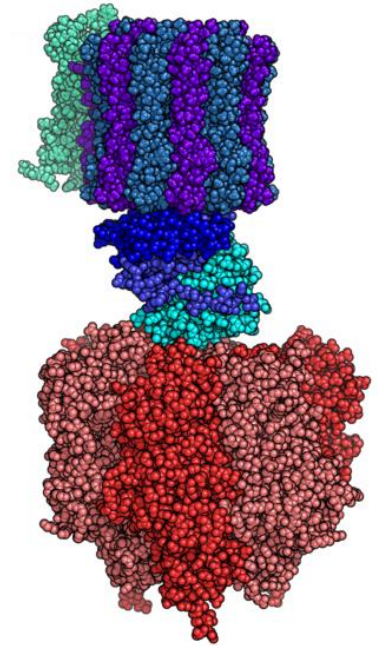
- Formação de estruturas supramacromoleculares



Ribossoma



Vírus

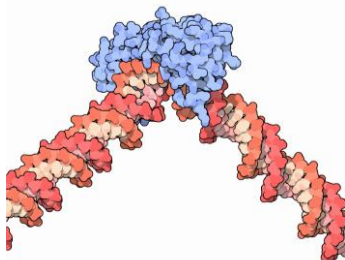
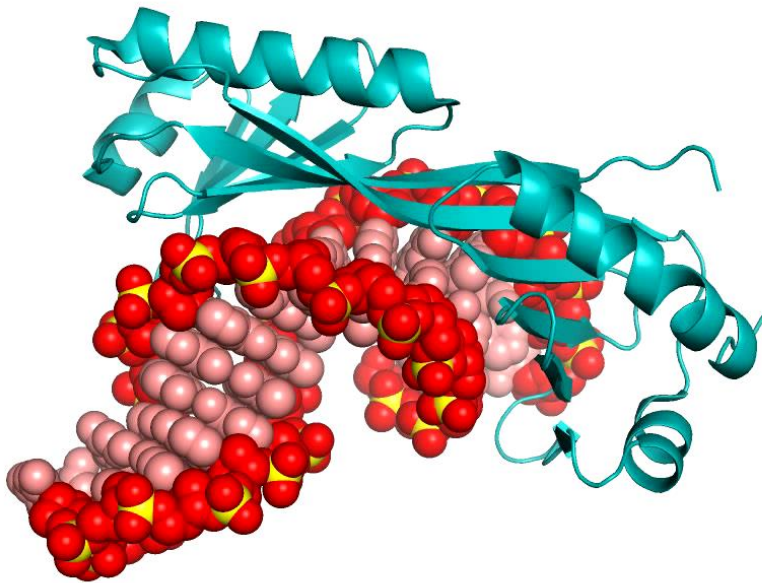


ATPase

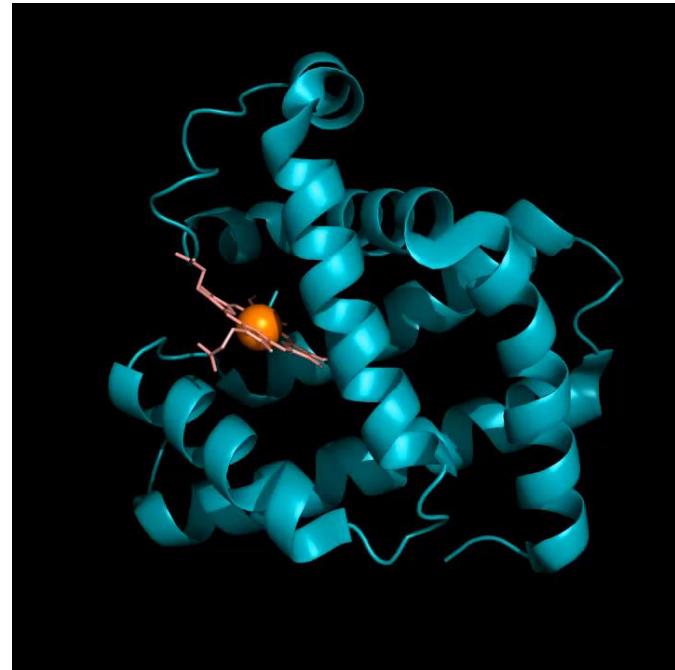
Função

Associação a ligandos

TATA-binding protein

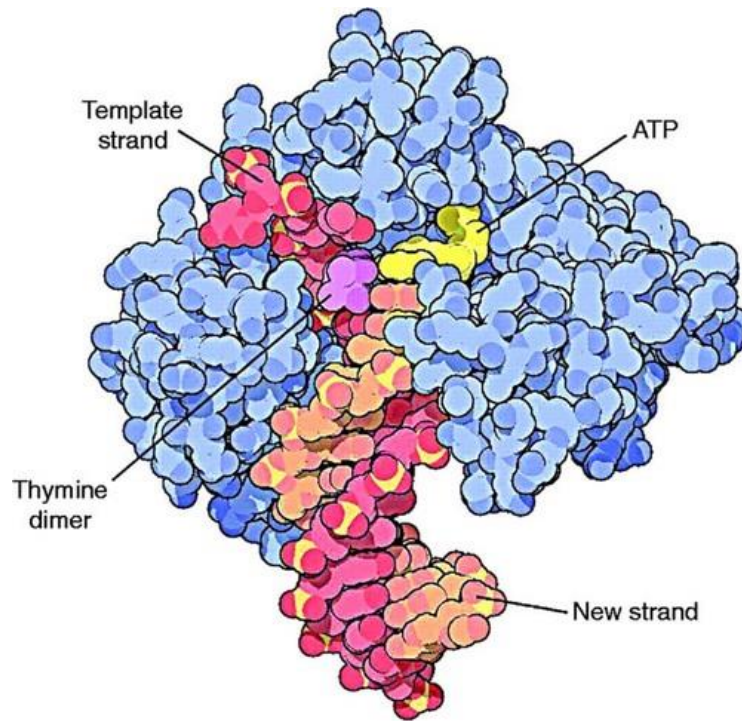


Myoglobin

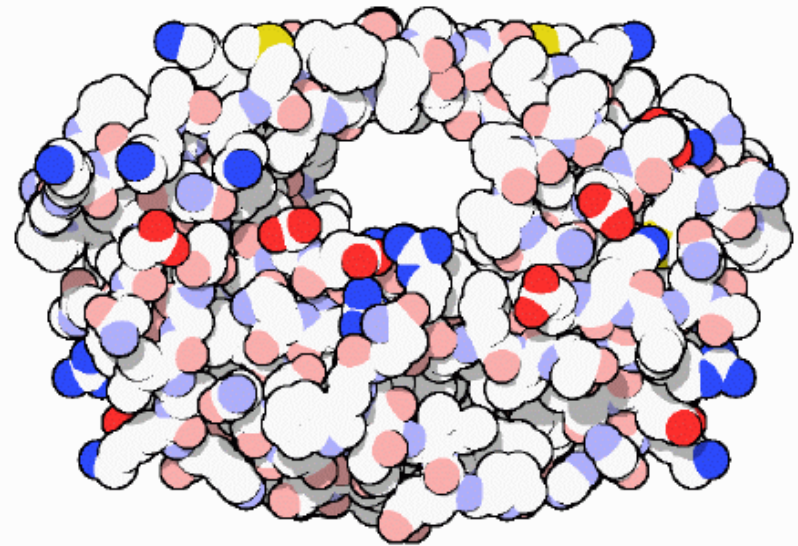


Catálise

DNA polymerase

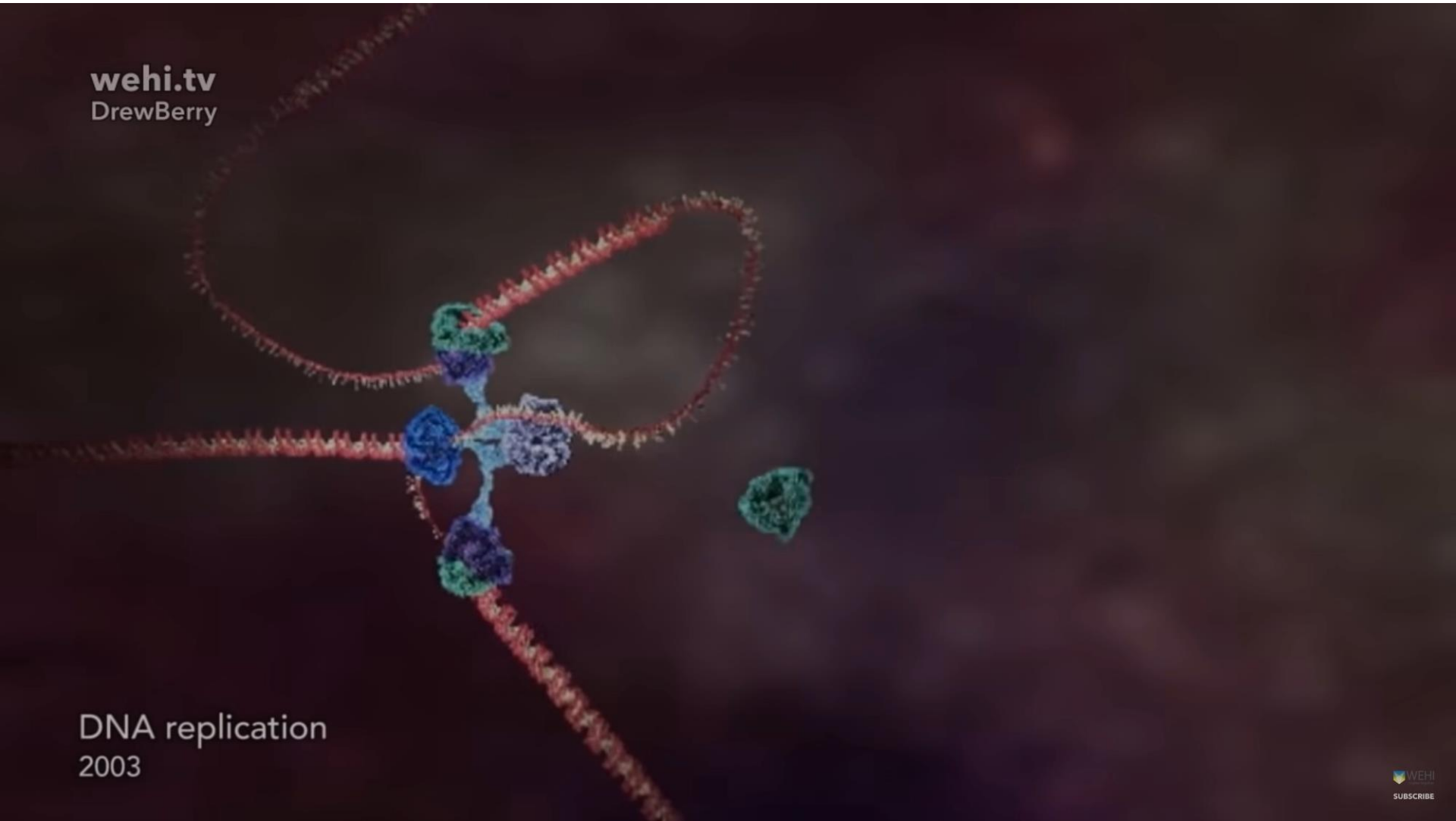


HIV protease



“Fotocopiadora” molecular

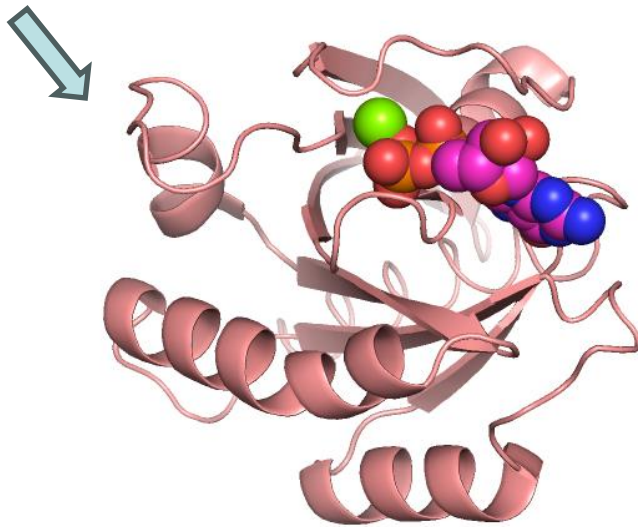
wehi.tv
DrewBerry



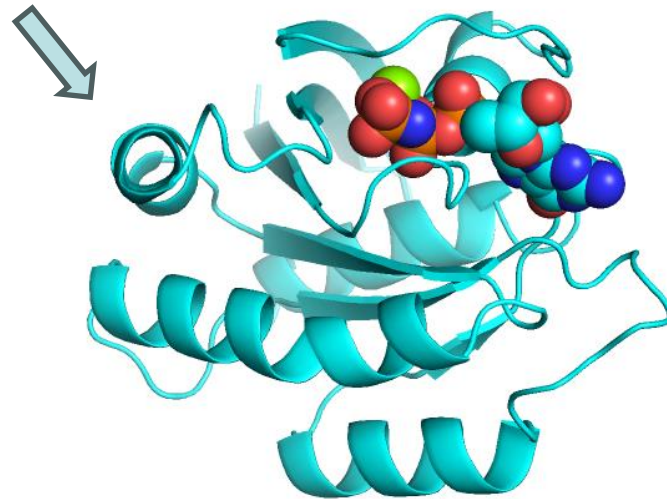
DNA replication
2003

Sinalização

ras protein

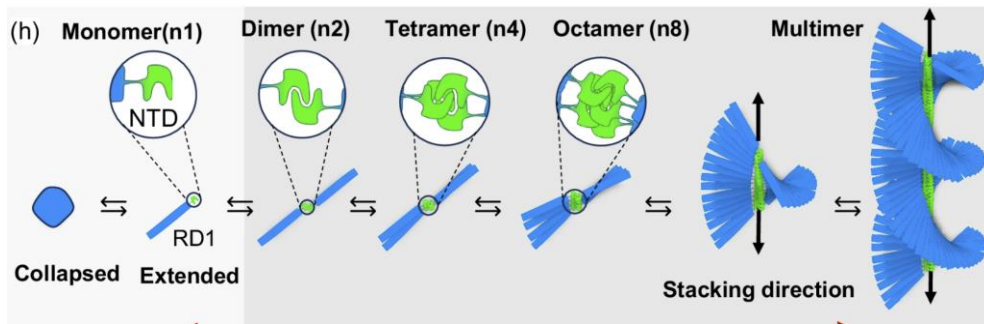
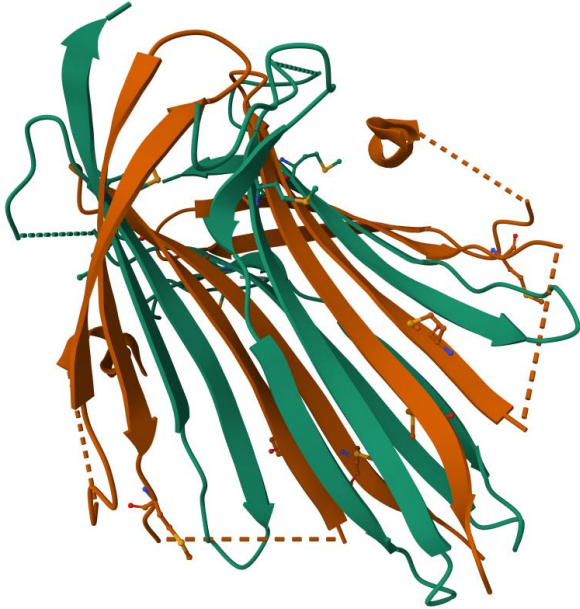


On

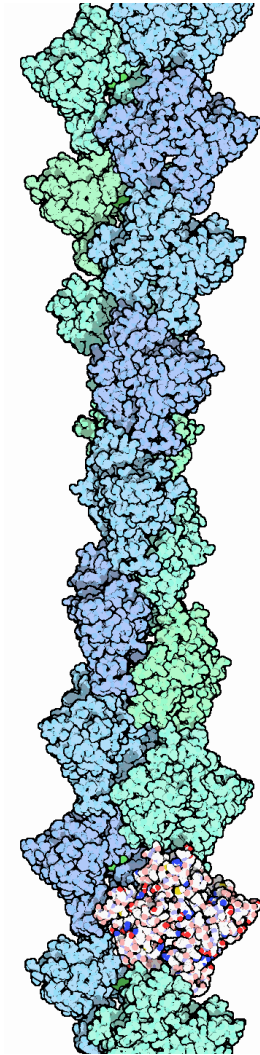


Off

Estrutura

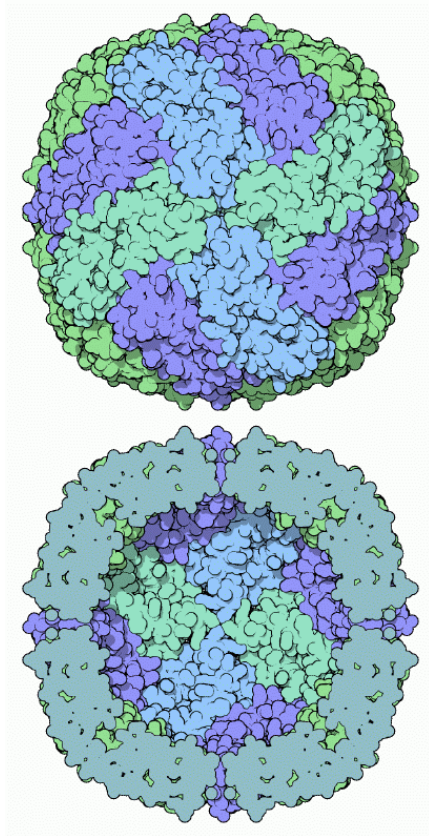


Silk fibroin

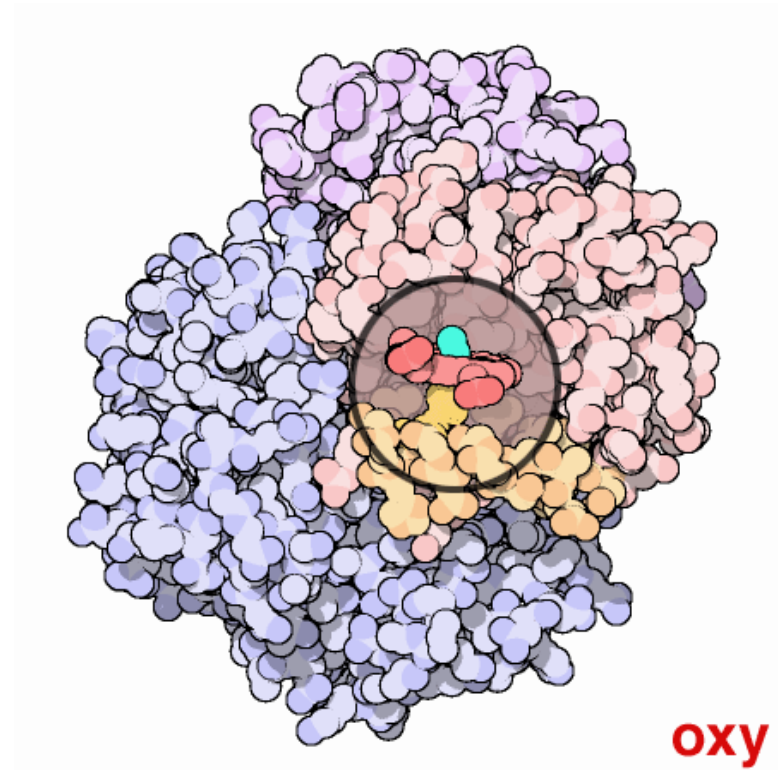


F-actin

Transporte

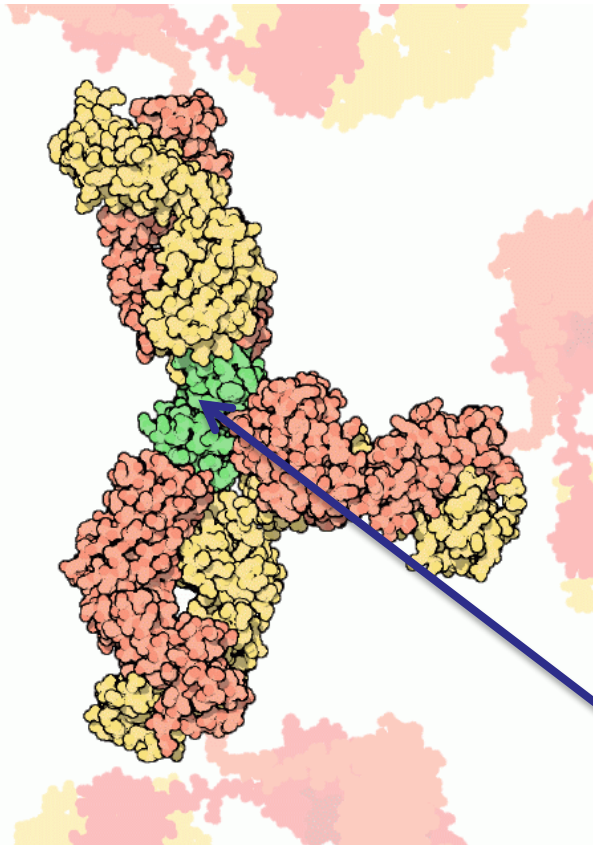


Ferritina (ferro)



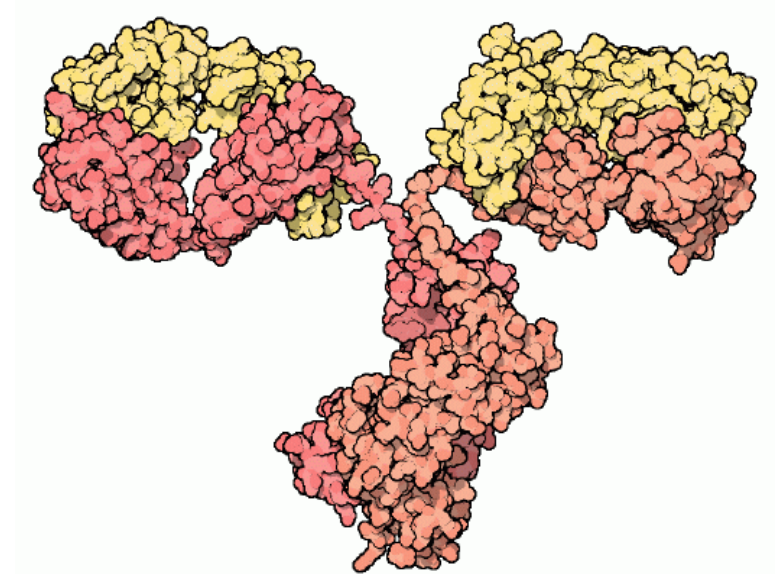
Hemoglobina (oxigénio)

Defesa

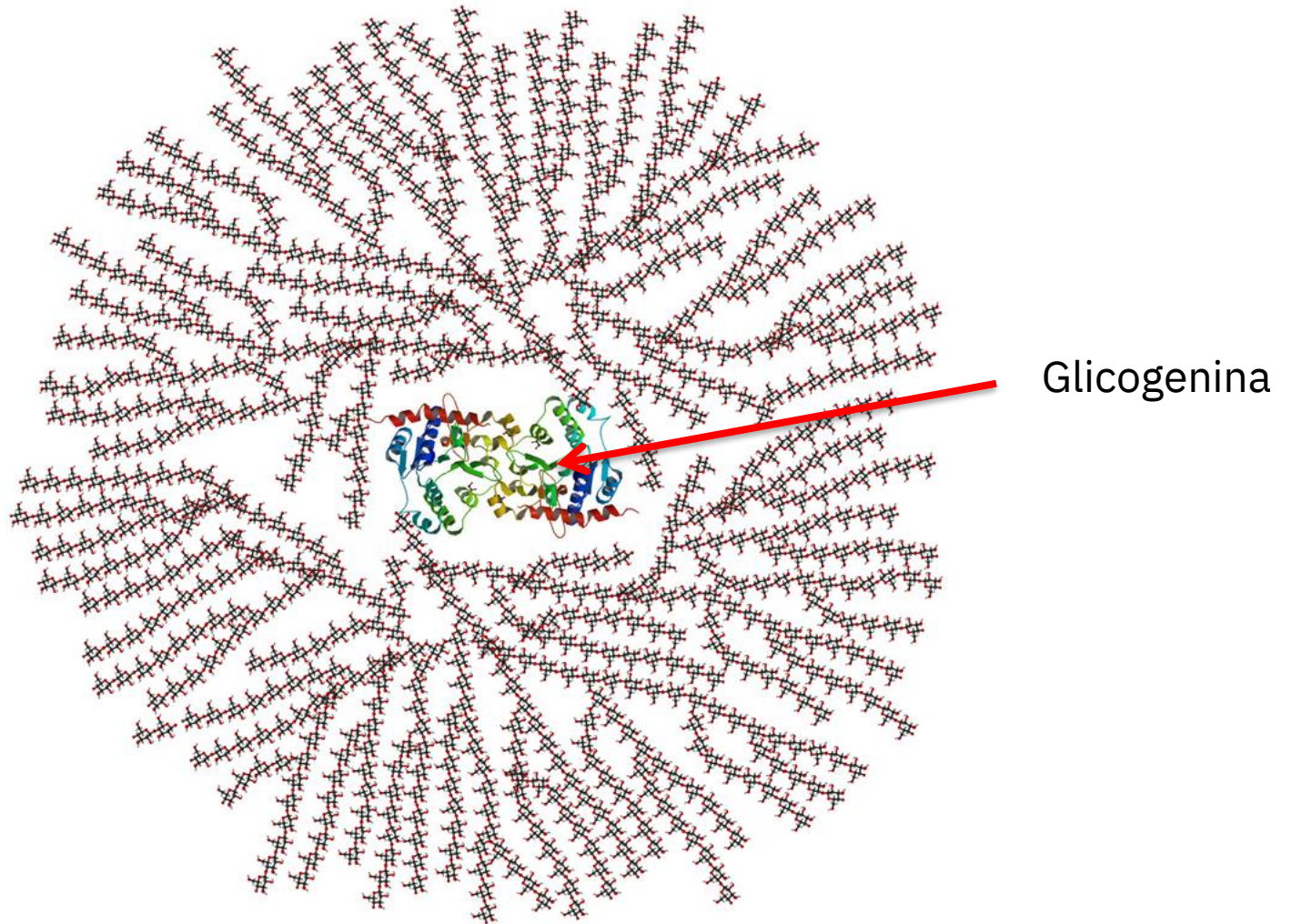


lisozima

imunoglobulina

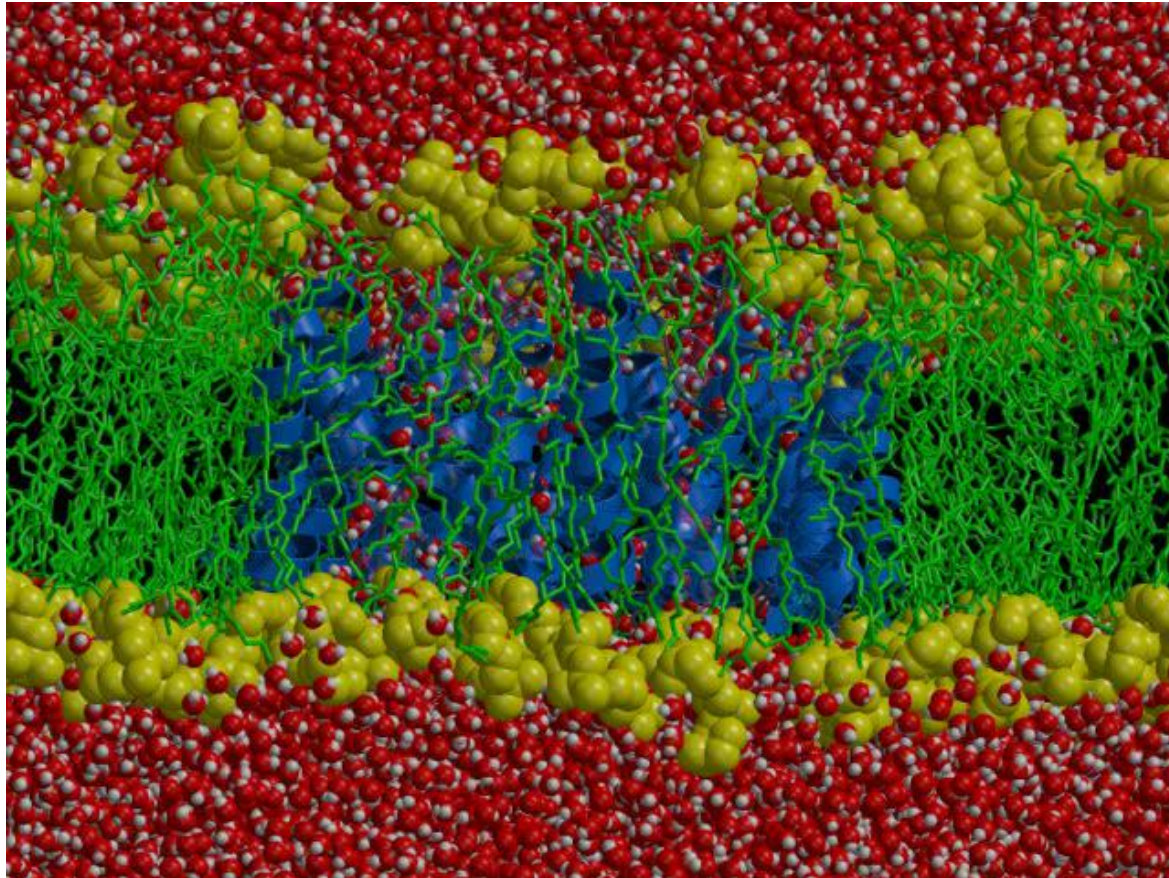


Armazenamento



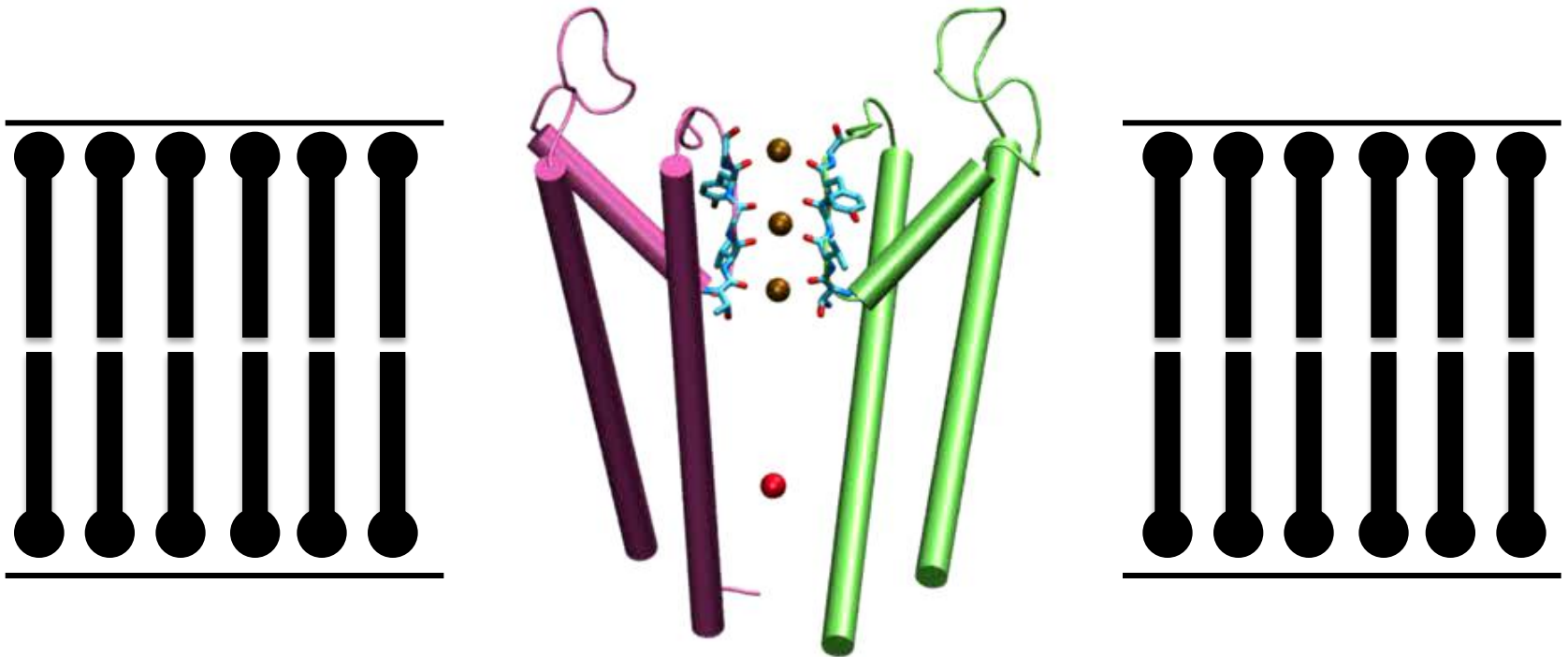
Glicogénio

Compartimentação



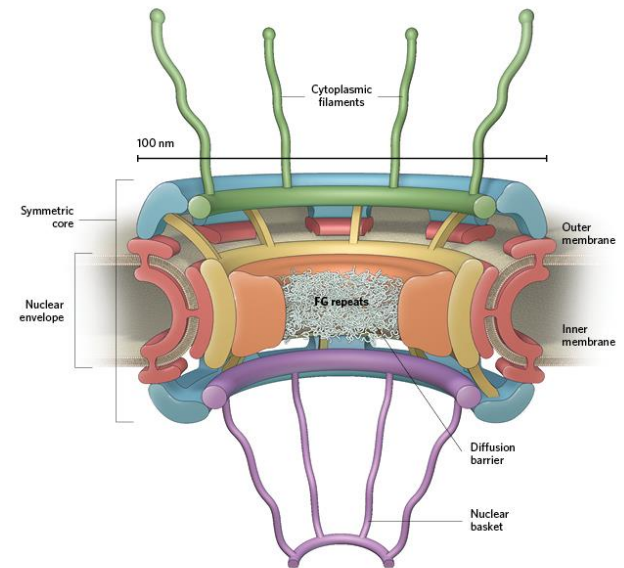
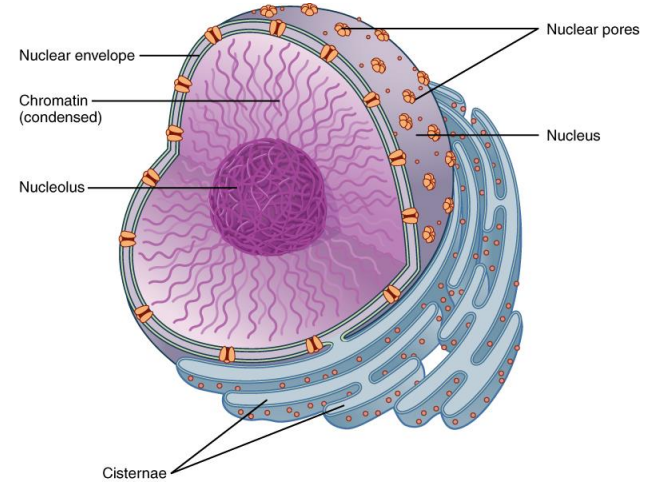
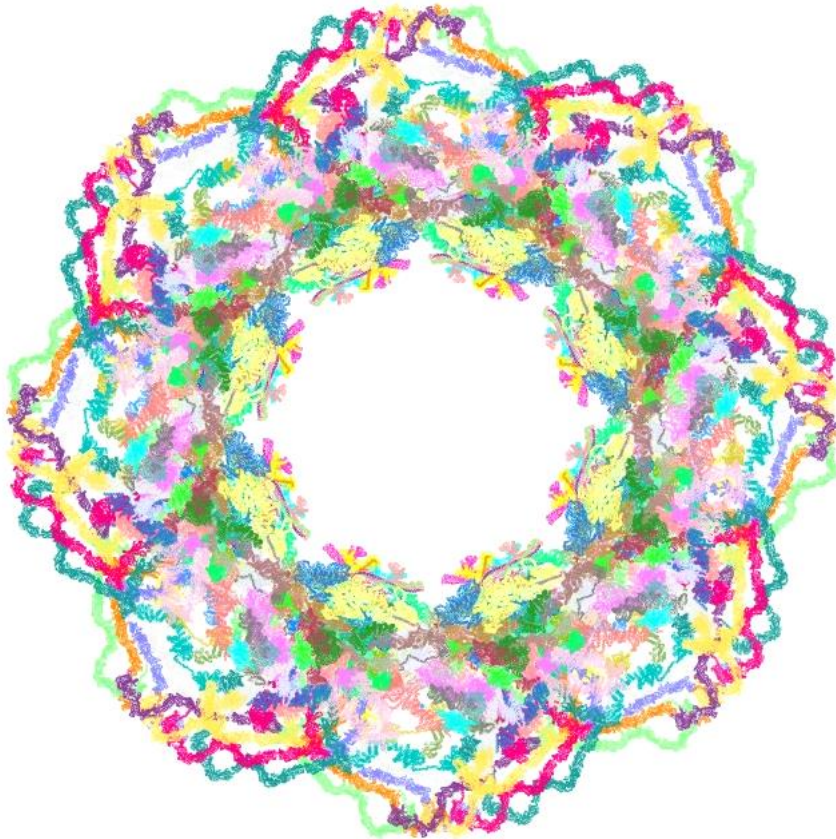
Membrana+aquaporina

Permeabilidade e transporte



As macromoléculas biológicas funcionam como máquinas moleculares

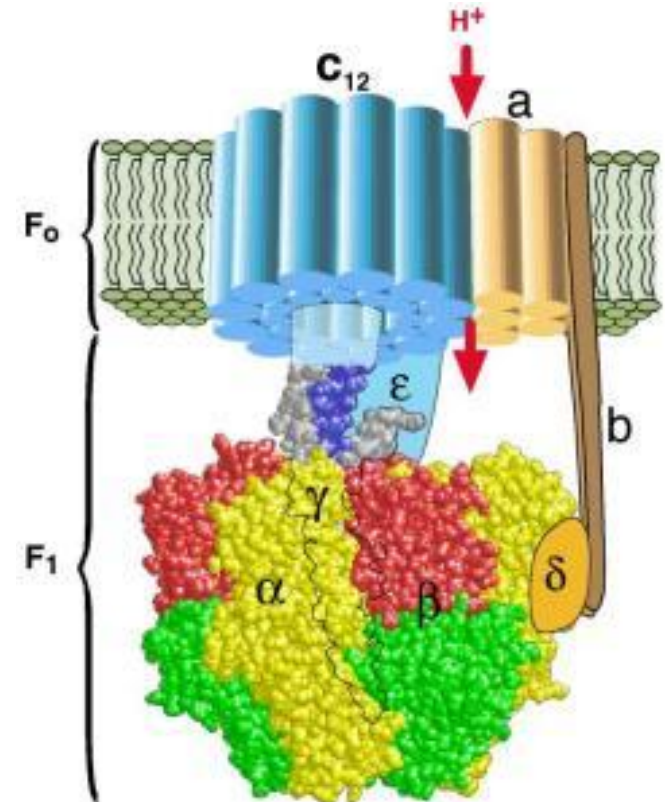
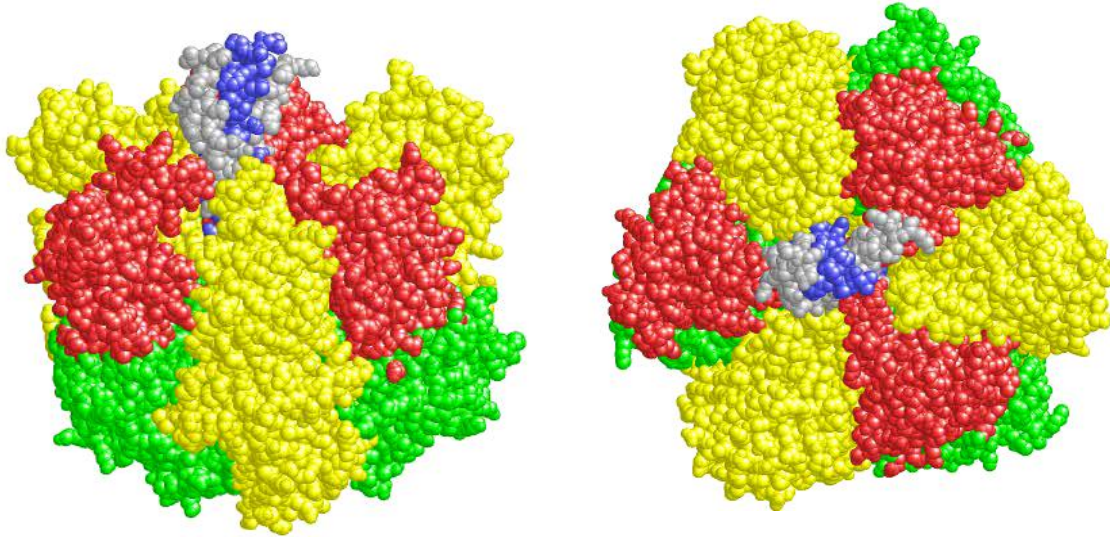
Estrutura de um poro nuclear



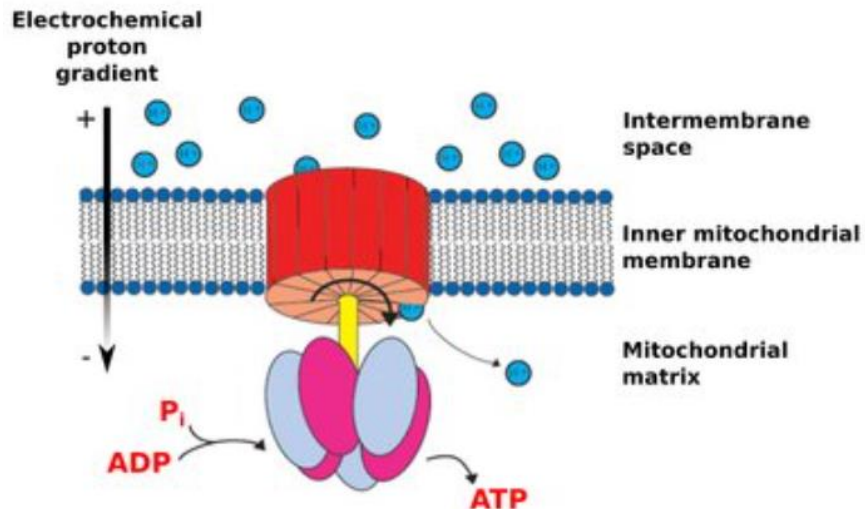
PDB: 7r5j

Conversão de energia

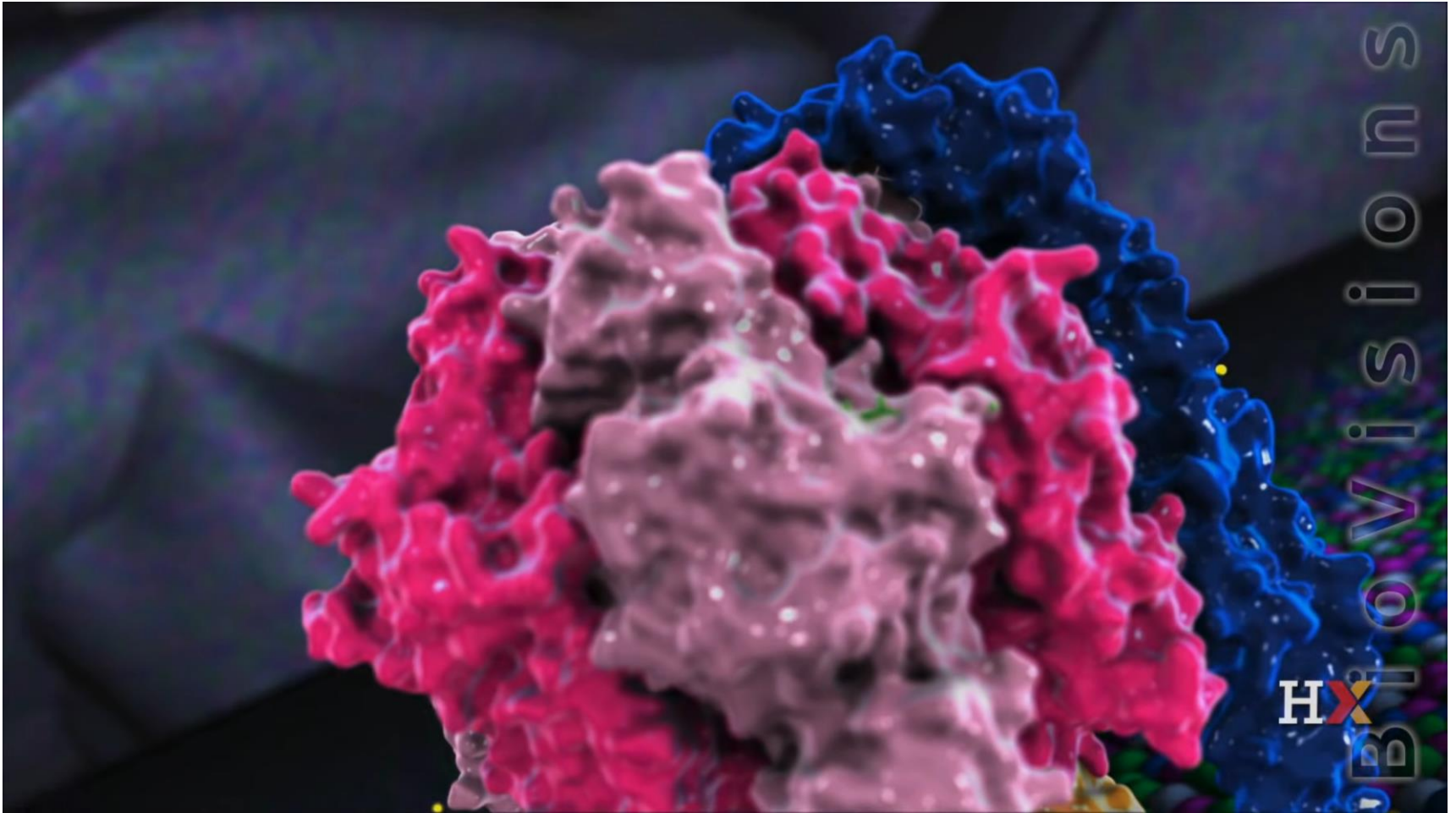
ATP sintase



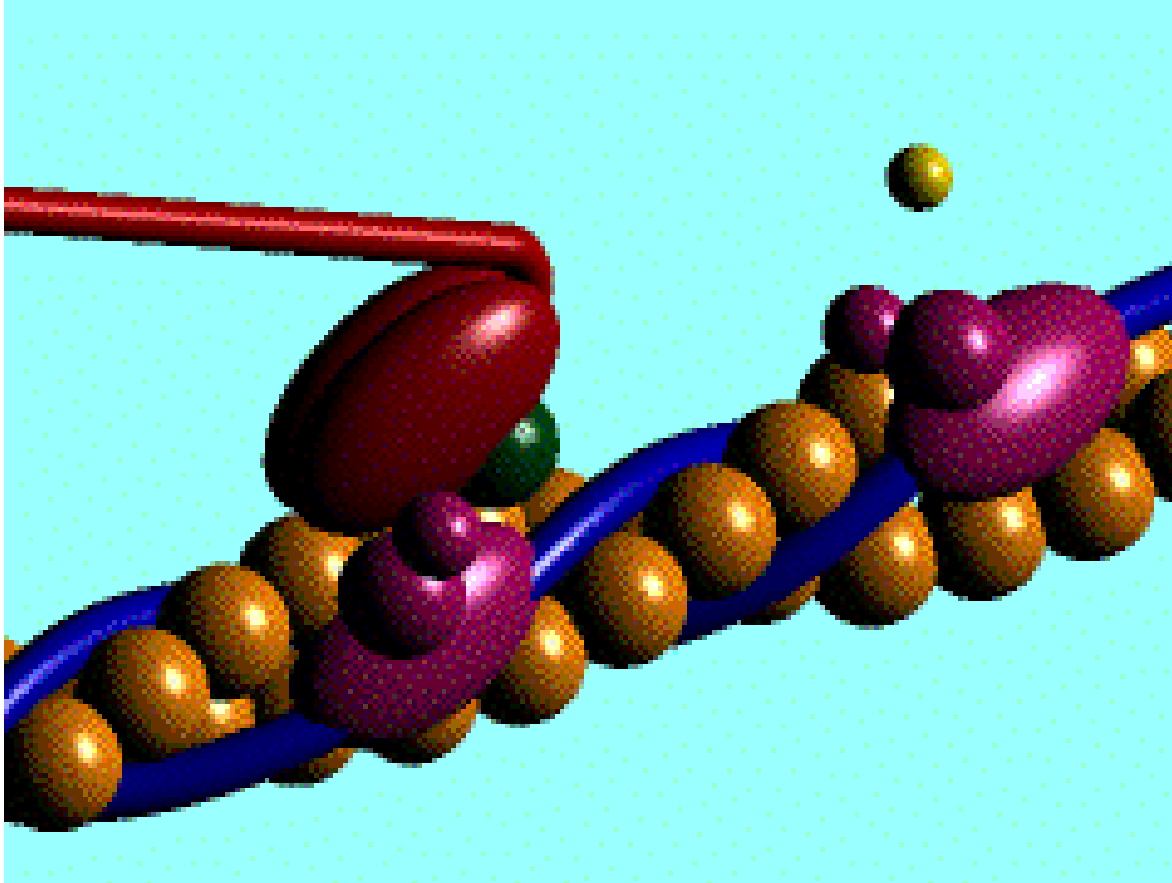
H. Wang and G. Oster (1998). Nature 396:279-282.



A molecular power generator

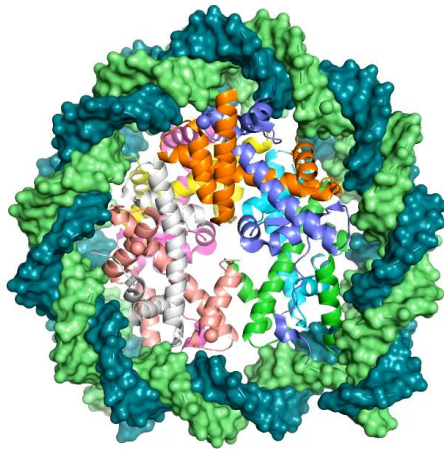
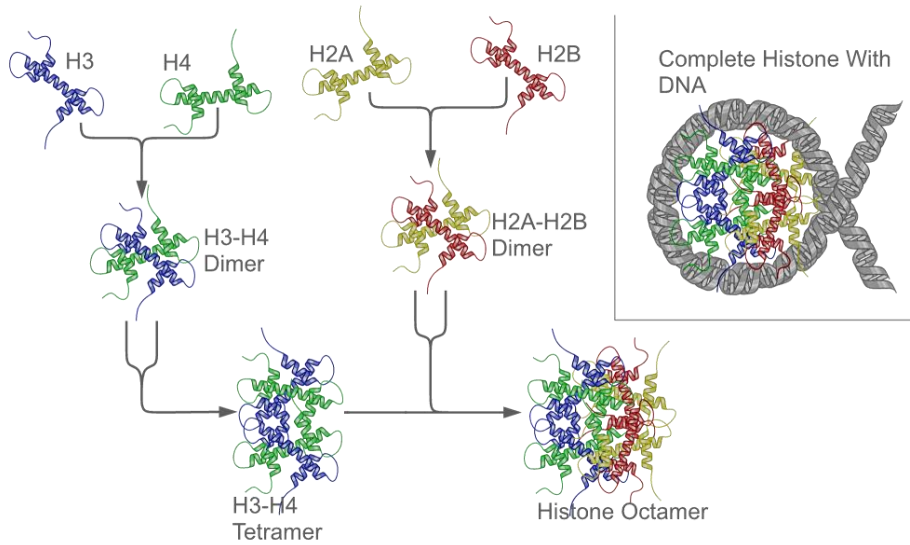


Motilidade

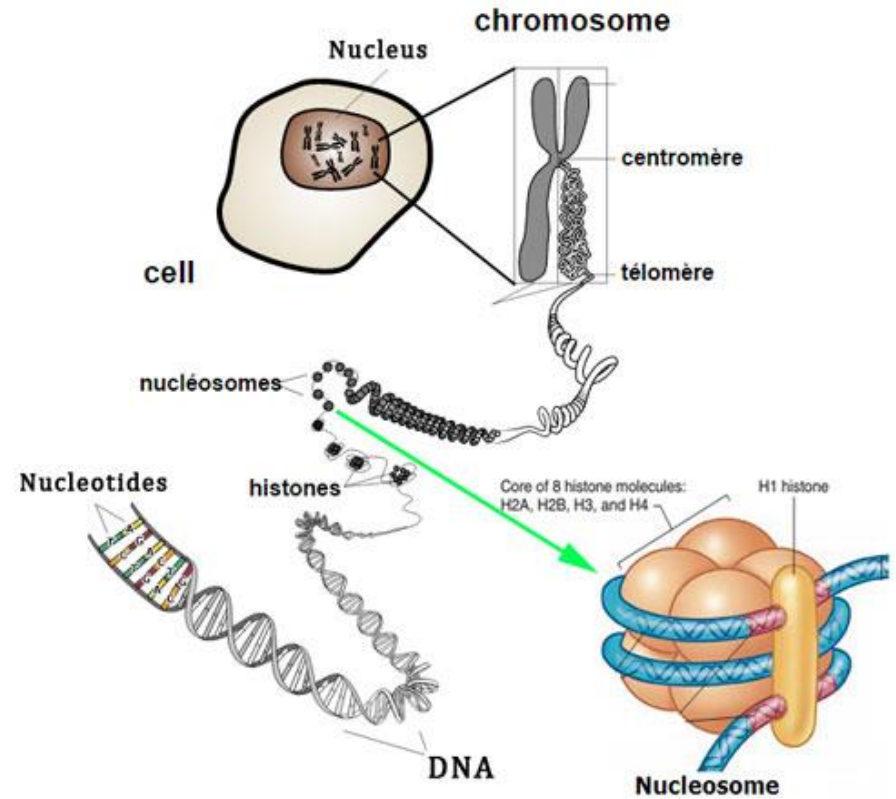


Actina+Miosina

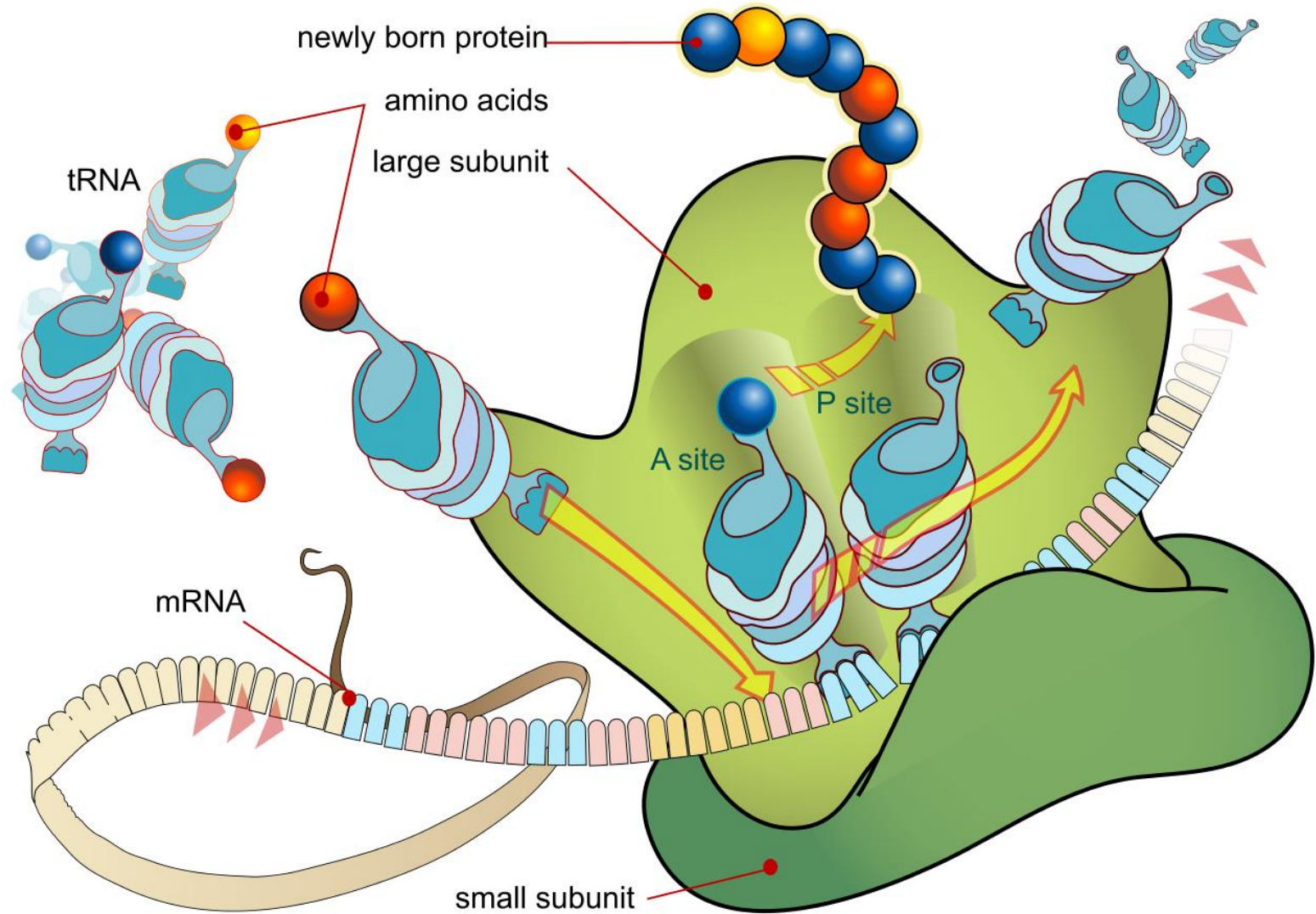
Informação



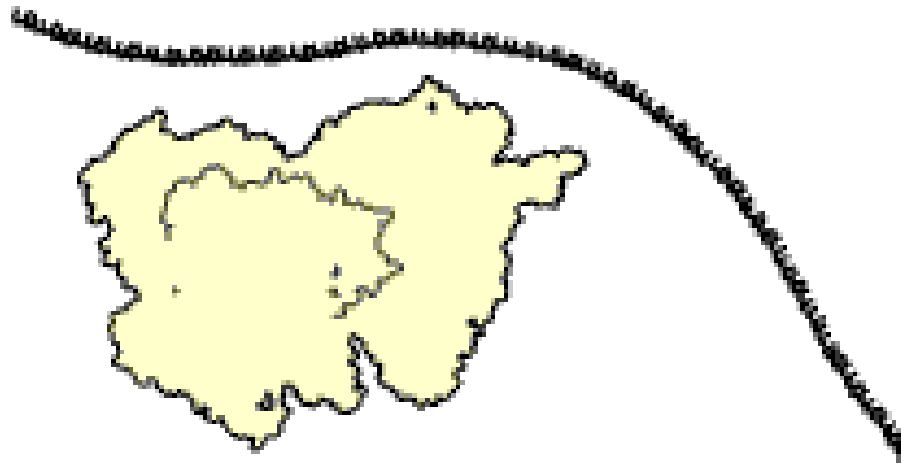
Nucleossoma



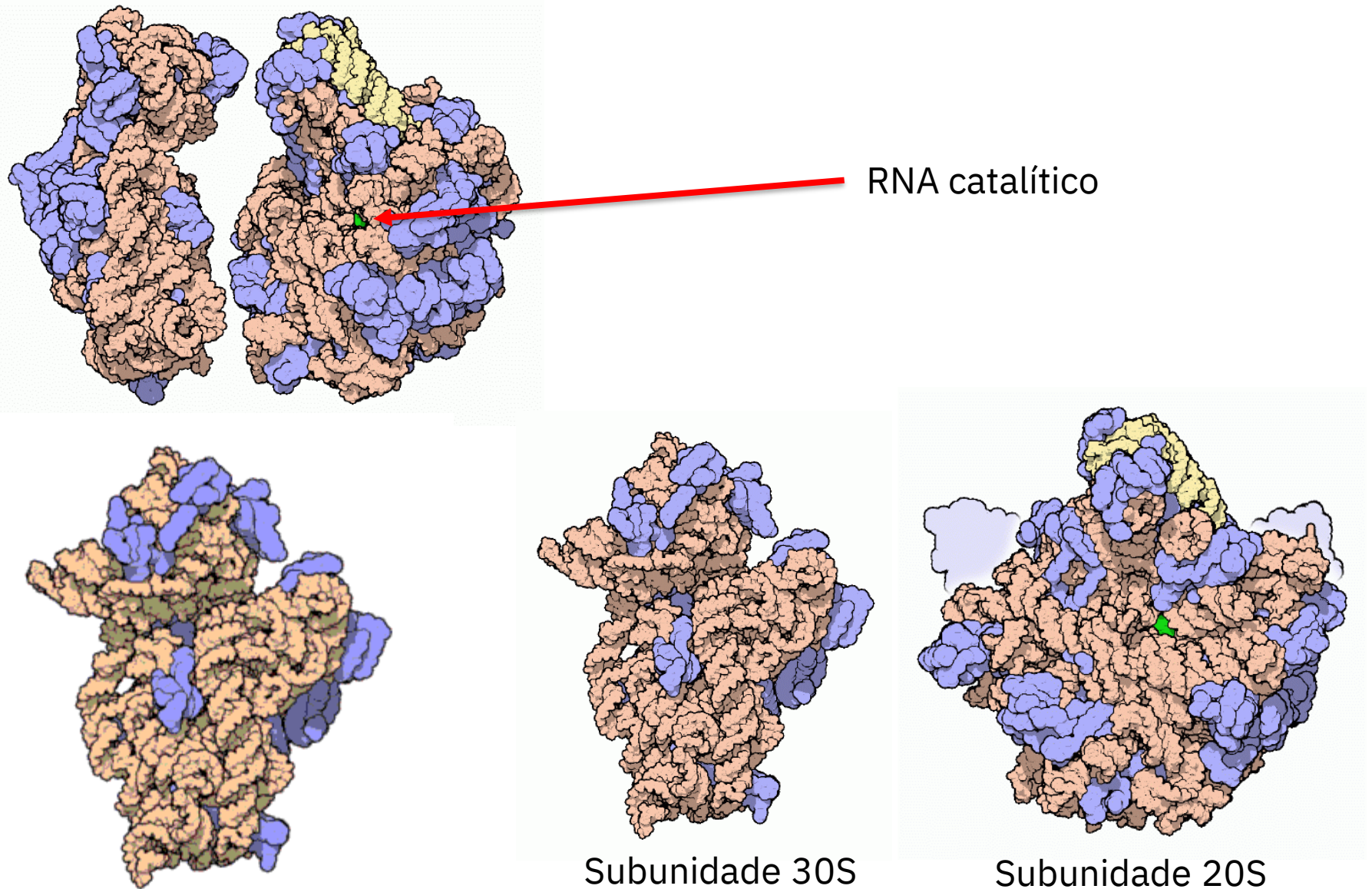
Síntese proteica



Síntese proteica (animação)

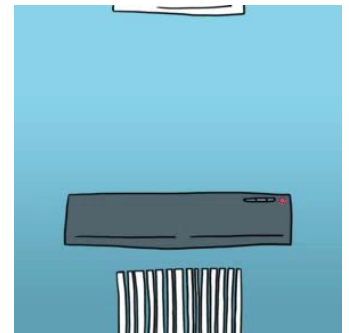
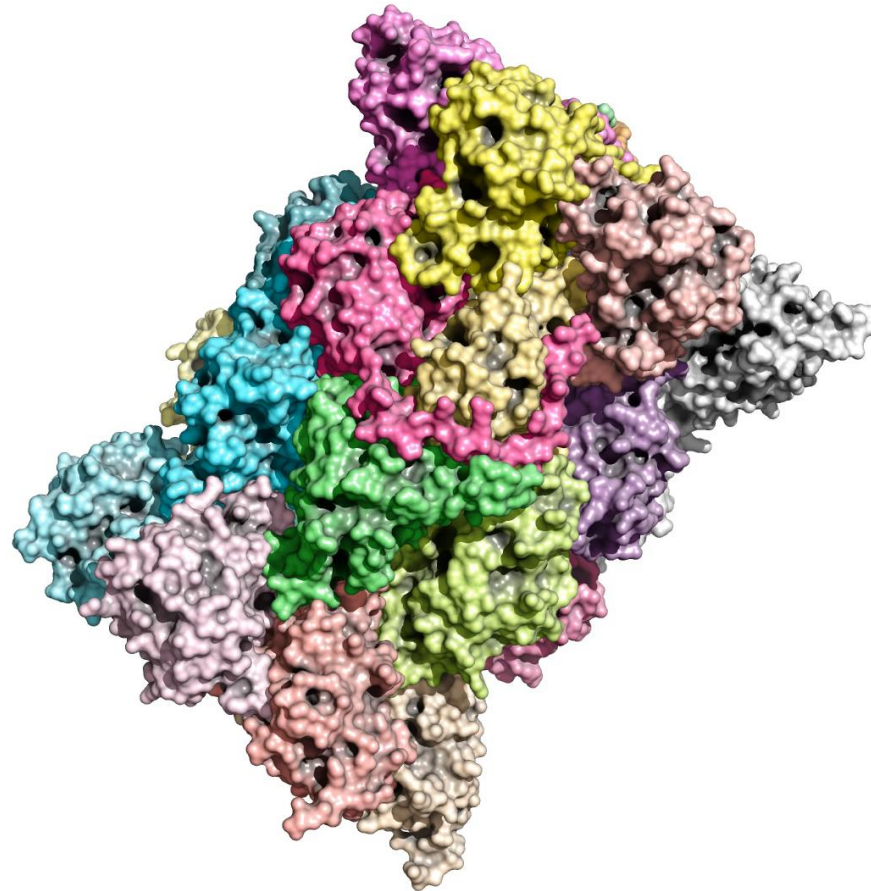


Estrutura do ribossoma bacteriano



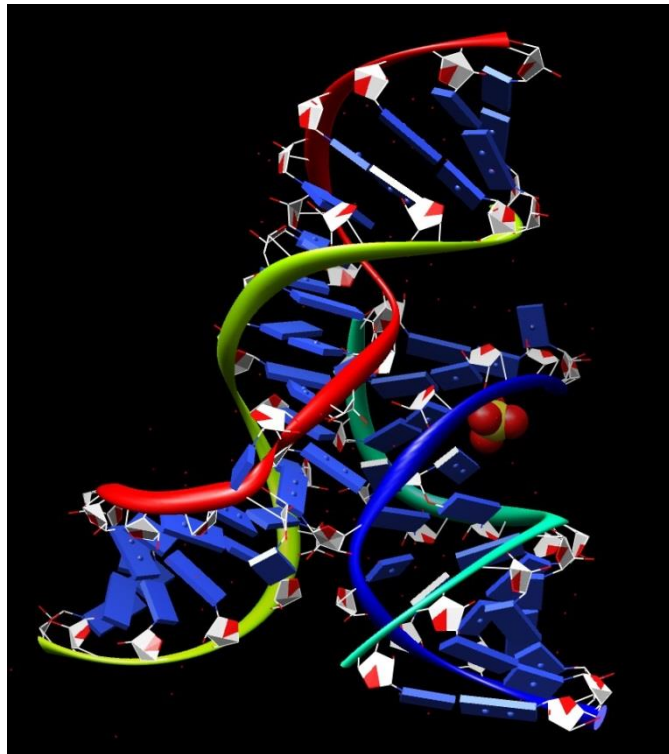
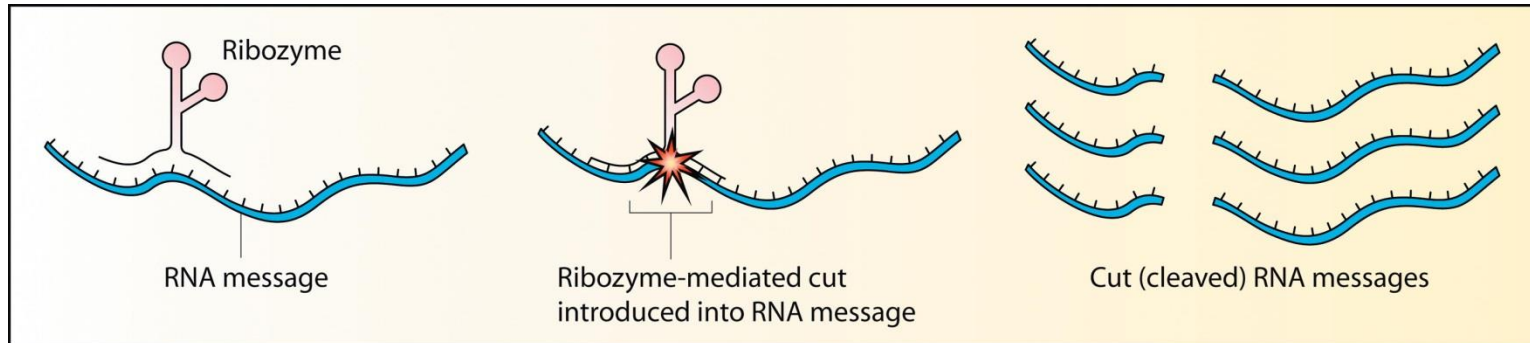
Reciclagem de proteínas

Proteassoma

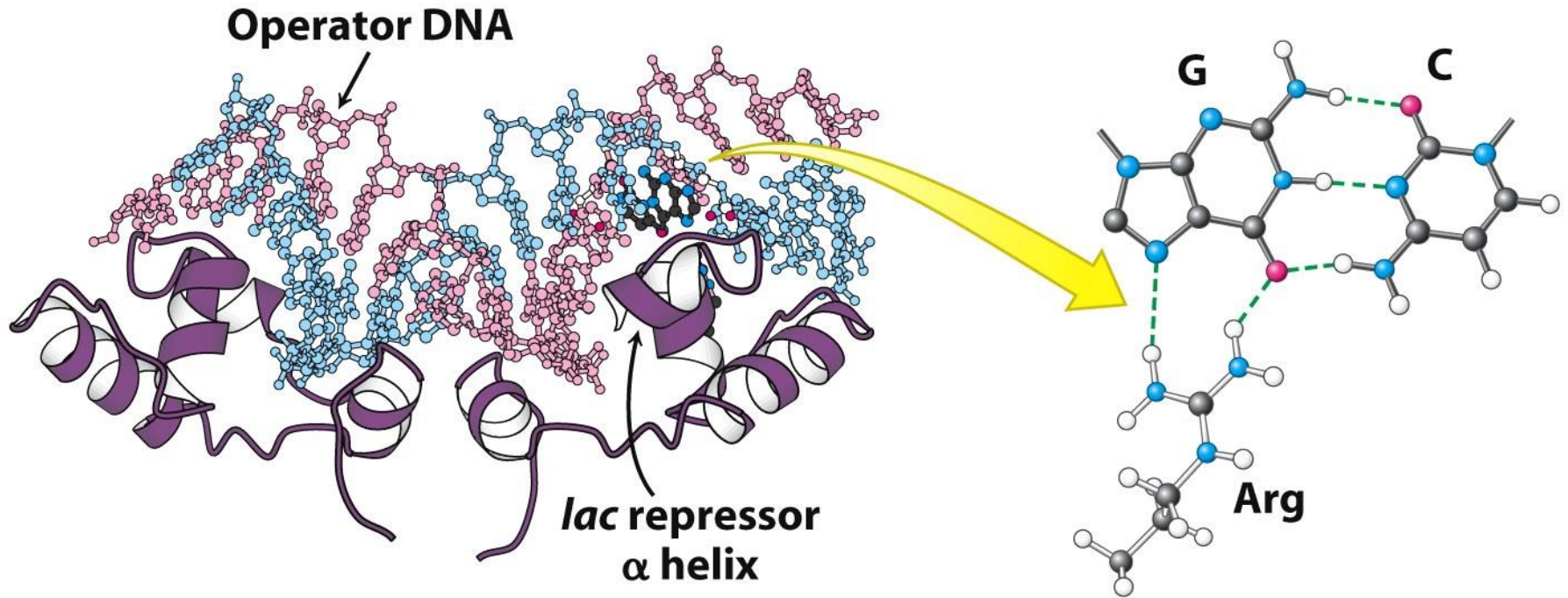


Assemblagem de máquinas moleculares

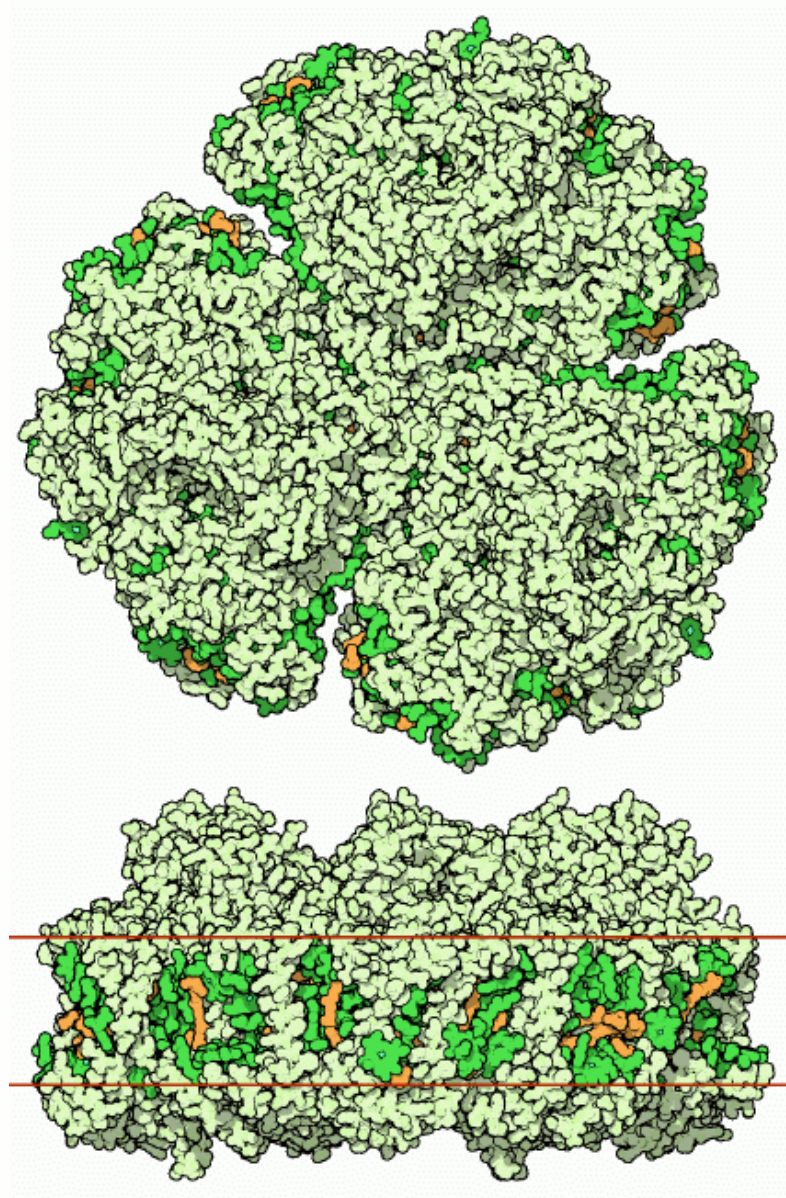
Ribozimas



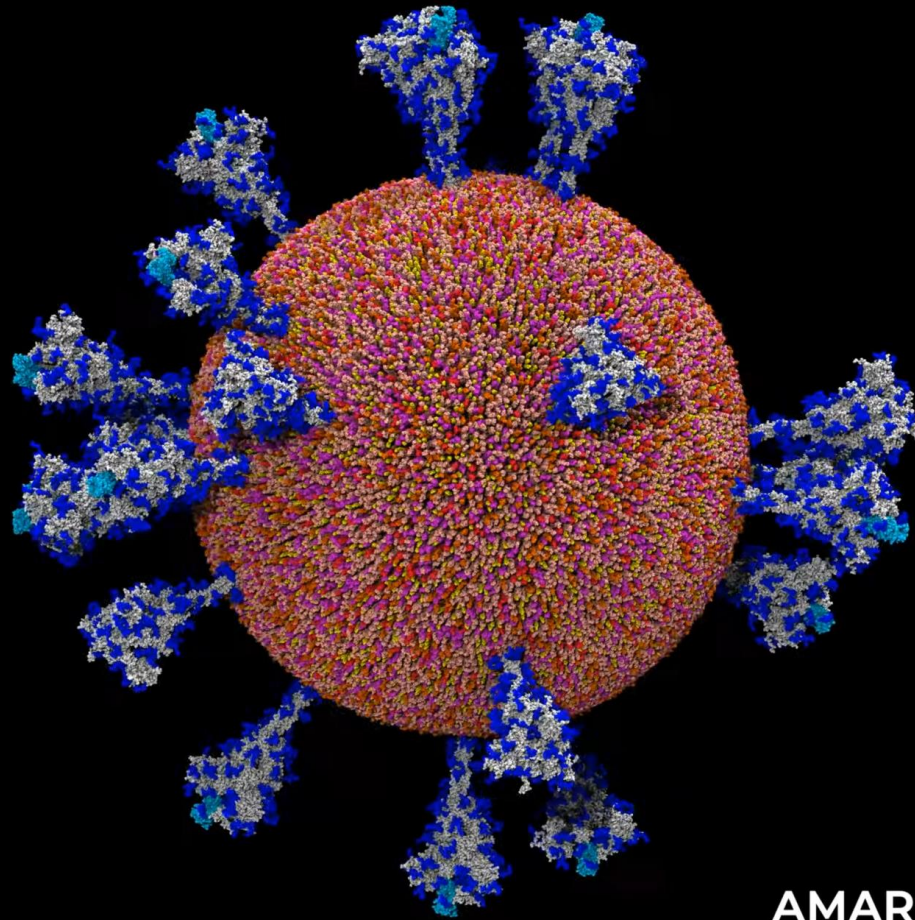
Regulação da transcrição



Fotossistema I



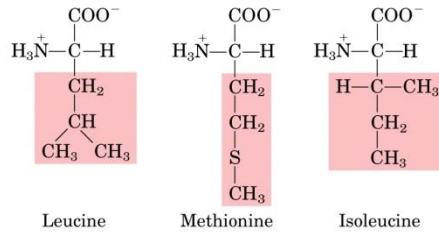
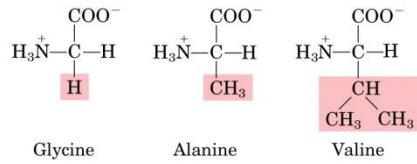
3.5M-atom model of SARS-CoV-2



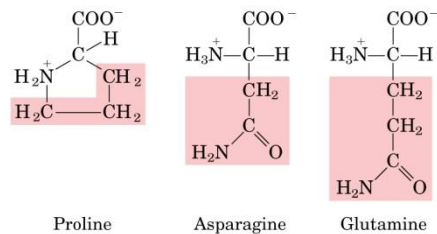
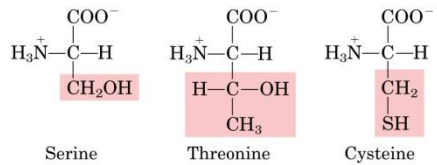
AMARO LAB - UCSD

Proteínas

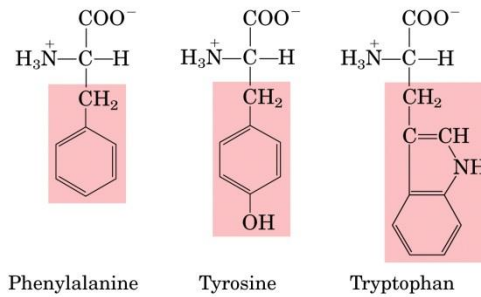
Nonpolar, aliphatic R groups



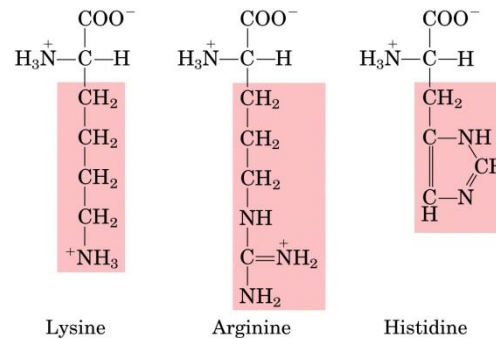
Polar, uncharged R groups



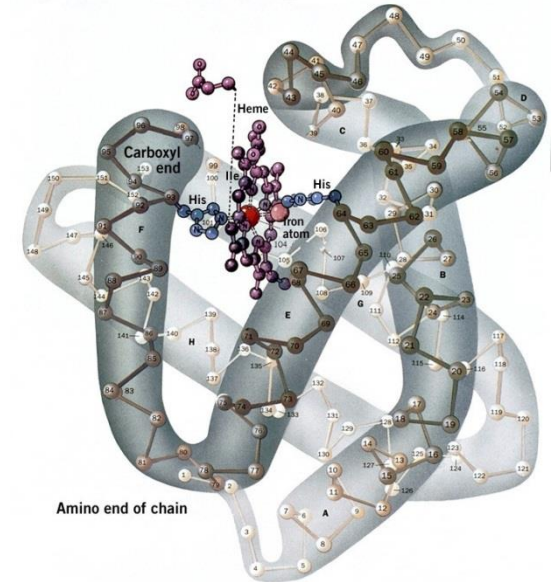
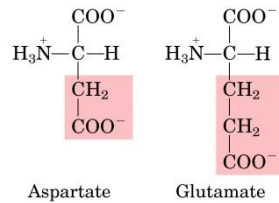
Aromatic R groups



Positively charged R groups



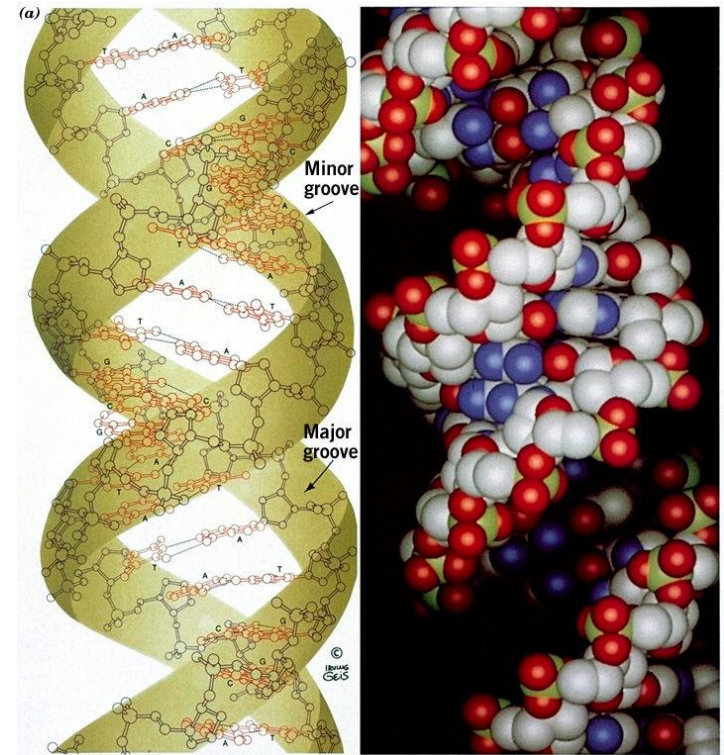
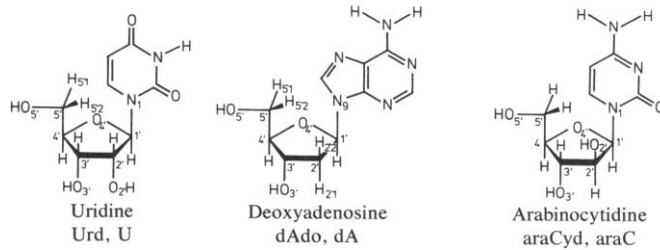
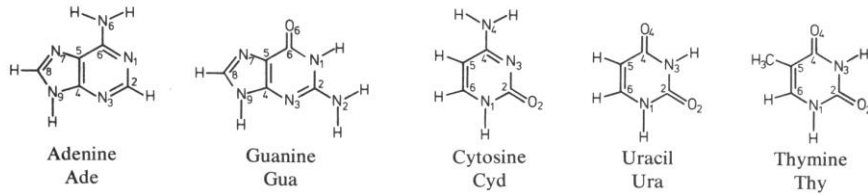
Negatively charged R groups



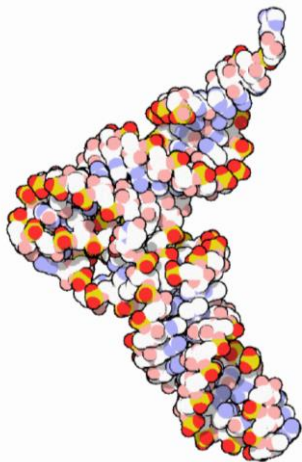
(a)

Myoglobin

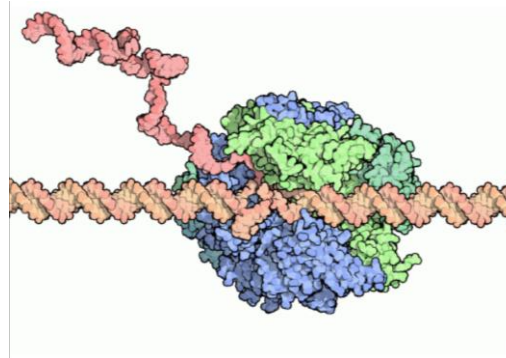
Ácidos nucleicos- blocos constituintes



B-Dna



RNA de transferência



RNA mensageiro